



LFCS

Linux Foundation Certified Sysadmin

Interview Questions & Answers

Complete Linux Sysadmin Preparation Guide

2025 EDITION

300

Questions

10

Topics

Beginner

to Advanced

Author: Akhilesh Chaudhari

Linux Foundation Certification Preparation Guide

LFCS Certification Preparation | 2025 Edition

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Beginner to Advanced

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PREFACE & INTRODUCTION

Welcome to the LFCS (Linux Foundation Certified Sysadmin) Interview Questions & Answers guide — your comprehensive companion for Linux Foundation Certification preparation.

The Linux Foundation Certified System Administrator (LFCS) certification validates the ability to perform basic to intermediate system administration tasks on Linux. It covers essential operations, user and group management, networking, storage, service management, and security.

This guide contains 300 carefully crafted interview questions and answers spanning all major LFCS topics, organized from beginner to advanced level. Whether you are preparing for the actual Linux Foundation examination or a technical interview for a Linux System Administrator role, this book will help you build strong foundational knowledge and advanced practical understanding.

Topics Covered: Essential System Operations & Commands • User & Group Management • Storage Management & Filesystems • Networking Fundamentals • Service Management & Security • Shell Scripting & Automation • System Monitoring & Troubleshooting • Advanced Topics & Virtualization • Kernel & Boot Management • Advanced Administration & Exam Prep

How to use this guide: Read through questions progressively from basic to advanced. Try to answer each question yourself before reading the provided answer. Use the questions as flashcards for revision. Focus on understanding concepts rather than memorizing answers.

Best of luck in your LFCS journey!

— Akhilesh Chaudhari

Section 1: Essential System Operations

Questions 1 – 30

Q1 What is the LFCS certification?

Answer : The Linux Foundation Certified System Administrator (LFCS) is a professional certification that validates skills in Linux system administration. It covers essential tasks such as managing files, configuring storage, installing software, managing users and services, and basic networking and security.

Q2 What is the Linux kernel?

Answer : The Linux kernel is the core of the Linux OS. It manages hardware resources (CPU, RAM, I/O devices) and provides a bridge between applications and hardware. It handles process scheduling, memory management, device drivers, and system calls.

Q3 How do you check the current Linux kernel version?

Answer : Use 'uname -r' to display the kernel release version. 'uname -a' shows all system information including kernel version, hostname, and hardware architecture. 'cat /proc/version' also provides detailed kernel version and build info.

Q4 What is systemd?

Answer : systemd is the modern init system and service manager for Linux. It is PID 1 – the first process started by the kernel. It manages system startup, service dependencies, logging (via journald), and targets (replacing runlevels). Key tool: systemctl.

Q5 How do you start, stop, and restart a service with systemd?

Answer : 'systemctl start servicename' starts a service immediately. 'systemctl stop servicename' stops it. 'systemctl restart servicename' stops and restarts it. 'systemctl reload servicename' reloads config without full restart. Always check status with 'systemctl status servicename'.

Q6 How do you enable a service to start at boot?

Answer : 'systemctl enable servicename' creates a symlink so the service starts at boot. 'systemctl disable servicename' removes it. 'systemctl enable --now servicename' both enables and starts it immediately. Use 'systemctl is-enabled servicename' to verify.

Q7 What is a systemd target?

Answer Targets are systemd's equivalent of SysV runlevels. Common targets: multi-user.target (runlevel 3 — multi-user text mode), graphical.target (runlevel 5 — GUI), rescue.target (runlevel 1 — single user), emergency.target (minimal shell). Use 'systemctl get-default' to see the default target.

Q8 How do you change the default systemd target?

Answer 'systemctl set-default multi-user.target' sets text-mode boot as default. 'systemctl set-default graphical.target' sets GUI boot. 'systemctl isolate rescue.target' switches to rescue mode immediately. These change /etc/systemd/system/default.target symlink.

Q9 What is journalctl and how do you use it?

Answer journalctl queries systemd's journal (structured log). Key uses: 'journalctl -u nginx' (service logs), 'journalctl -f' (follow live), 'journalctl -p err' (errors only), 'journalctl --since today' (today's logs), 'journalctl --since "2025-01-01" --until "2025-01-31"' (date range), 'journalctl -b' (current boot).

Q10 How do you view and set the hostname?

Answer 'hostname' displays the current hostname. 'hostnamectl' shows detailed hostname info. 'hostnamectl set-hostname newhostname' permanently changes it. It updates /etc/hostname and /etc/hosts. Always verify with 'hostnamectl status' after changing.

Q11 What is the /proc filesystem?

Answer /proc is a virtual filesystem (procfs) that exposes runtime kernel and process information. Key files: /proc/cpuinfo (CPU details), /proc/meminfo (memory stats), /proc/loadavg (system load), /proc/PID/ (per-process info), /proc/sys/ (kernel tunables). Not a real disk filesystem — data is generated dynamically.

Q12 How do you manage kernel parameters with sysctl?

Answer 'sysctl -a' lists all kernel parameters. 'sysctl vm.swappiness' shows a specific value. 'sysctl -w vm.swappiness=10' changes it temporarily. For persistence, add 'vm.swappiness=10' to /etc/sysctl.conf or files in /etc/sysctl.d/, then run 'sysctl -p' to reload.

Q13 What are runlevels and how do they map to systemd targets?

Answer Runlevel 0=poweroff.target, 1=rescue.target, 2/3/4=multi-user.target, 5=graphical.target, 6=reboot.target. In systemd, 'systemctl isolate multi-user.target' switches runlevel. Legacy 'init 3' still works on most systems as a compatibility shim.

Q14 What is the purpose of `/etc/os-release`?

Answer `/etc/os-release` contains operating system identification data in shell-compatible key=value format. Fields include NAME (distro name), VERSION, ID (short identifier like 'ubuntu'), PRETTY_NAME, and VERSION_ID. Used by scripts and tools to detect the Linux distribution and version without parsing complex output.

Q15 How do you display system resource usage?

Answer `top/htop`: interactive CPU and memory monitor. `'vmstat 1 5'`: virtual memory stats every second, 5 times. `'iostat -x 1'`: disk I/O statistics. `'free -h'`: memory and swap usage. `'uptime'`: system uptime and load average. `'sar'` (sysstat package): historical performance data collection and reporting.

Q16 What is the load average in Linux?

Answer Load average represents the average number of processes in the run queue (running or waiting for CPU) over 1, 5, and 15 minute intervals. Shown by `uptime` and `top`. A load of 1.0 per CPU core is 100% utilized. Load > number of cores indicates the system is overloaded.

Q17 How do you find which process is using the most CPU?

Answer Run `'top'` and press 'P' to sort by CPU. `'ps aux --sort=-%cpu | head -20'` lists top CPU consumers. `'pidstat 1'` from sysstat shows per-process CPU. `'htop'` provides a color-coded interactive view. `'ps -eo pid,ppid,cmd,%cpu,%mem --sort=-%cpu | head'` also works.

Q18 What is the purpose of the `/etc` directory?

Answer `/etc` contains system-wide configuration files. Key files: `/etc/passwd` (users), `/etc/shadow` (password hashes), `/etc/fstab` (filesystems), `/etc/hosts` (hostname resolution), `/etc/resolv.conf` (DNS), `/etc/ssh/` (SSH config), `/etc/cron.d/` (cron jobs). Config changes here affect the whole system.

Q19 What is the difference between hard and soft (symbolic) links?

Answer A hard link is a directory entry pointing to the same inode — same data, same permissions, not broken if original deleted. A symbolic (soft) link is a file that points to another file by path — like a shortcut. If the target is deleted, the symlink is broken. Hard links can't span filesystems; symlinks can.

Q20 How do you create hard and symbolic links?

Answer 'ln file hardlink' creates a hard link. 'ln -s target symlink' creates a symbolic link. Verify with 'ls -li' (hard links share the same inode number) or 'ls -la' (symlinks show arrow '->'). Use 'readlink -f symlink' to resolve the full path of a symlink.

Q21 How do you archive and compress files?

Answer tar is the primary archiving tool. 'tar -czf archive.tar.gz /path' creates gzip-compressed archive. 'tar -cjf archive.tar.bz2 /path' uses bzip2. 'tar -xzf archive.tar.gz' extracts. 'tar -tf archive.tar.gz' lists contents without extracting. For just compression: gzip, bzip2, xz commands.

Q22 What does the file command do?

Answer 'file filename' determines the file type by examining its magic bytes (content), not the extension. Useful when extensions are missing or misleading. Examples: 'file script.sh' returns 'Bourne-Again shell script', 'file image.jpg' returns JPEG image info. Works on binaries, text, archives, and more.

Q23 How do you check system memory usage?

Answer 'free -h' shows RAM and swap in human-readable format. 'cat /proc/meminfo' provides detailed memory breakdown. 'vmstat -s' shows memory stats. In top/htop, memory columns show total/used/free. 'smem' shows per-process memory. Key values: MemTotal, MemFree, MemAvailable (best indicator of usable memory).

Q24 What is a process and how do you list running processes?

Answer A process is a running instance of a program with its own PID, memory space, and resources. 'ps aux' lists all processes with user, CPU%, memory%, and command. 'ps -ef' shows parent PID (PPID). 'pgrep nginx' finds PIDs by name. 'ls /proc/' also lists all PIDs as directories.

Q25 How do you send signals to processes?

Answer 'kill PID' sends SIGTERM (15) — graceful stop. 'kill -9 PID' sends SIGKILL — force terminate. 'kill -HUP PID' sends SIGHUP — reload config. 'killall processname' kills by name. 'pkill -u username' kills all processes by user. 'kill -l' lists all signal names and numbers.

Q26 What is the difference between foreground and background processes?

Answer Foreground processes run in the terminal and block it until done. Background processes run without blocking the terminal. '&' appends to a command to run it in background ('sleep 100 &'). 'jobs' lists background jobs. 'fg %1' brings job 1 to foreground. Ctrl+Z suspends a foreground process.

Q27 What is the nice and renice command?

Answer 'nice -n 10 command' starts a process with priority +10 (lower priority, ranges -20 to 19).
:
Higher nice = lower priority. 'renice -n 5 -p PID' adjusts priority of a running process. Only root can set negative nice values (higher priority). 'ps -eo pid,ni,cmd' shows current nice values.

Q28 How do you schedule one-time tasks in Linux?

Answer 'at' schedules a one-time job. 'echo "backup.sh" | at 3am tomorrow' schedules a backup.
:
'atq' lists pending at jobs. 'atrm job_number' removes a job. The atd daemon must be running. 'at now + 2 hours' runs 2 hours from now. Access controlled via /etc/at.allow and /etc/at.deny.

Q29 How do you view and manage boot messages?

Answer 'dmesg' shows kernel ring buffer (hardware detection, driver messages, boot errors).
:
'dmesg -T' adds human-readable timestamps. 'dmesg | grep -i error' finds errors. 'journalctl -b' shows all logs from current boot. 'journalctl -b -1' shows previous boot. 'journalctl --list-boots' shows all recorded boots.

Q30 What is the purpose of /etc/environment?

Answer /etc/environment is a PAM-read file that sets system-wide environment variables for all users, regardless of shell. Unlike .bashrc or .profile, it is not a shell script — just KEY=VALUE pairs. Used for settings like PATH, JAVA_HOME, or LANG that should apply globally. Requires re-login to take effect.

Section 2: User & Group Management

Questions 31 – 60

Q31 How do you create a new user in Linux?

Answer 'useradd -m -s /bin/bash -G sudo john' creates user john with home directory, bash shell, and sudo group. '-m' creates home dir, '-s' sets shell, '-G' adds supplementary groups, '-d' specifies custom home path, '-u' specifies UID. 'adduser john' on Debian/Ubuntu provides an interactive, friendlier version.

Q32 How do you modify an existing user account?

Answer 'usermod' modifies user properties. 'usermod -s /bin/zsh john' changes shell. 'usermod -aG docker john' adds john to docker group (use -a to append, not replace). 'usermod -L john' locks account. 'usermod -U john' unlocks. 'usermod -e 2025-12-31 john' sets account expiry date.

Q33 How do you delete a user and their files?

Answer 'userdel john' removes the user but keeps home directory. 'userdel -r john' removes user, home directory, and mail spool. Check for running processes first: 'ps -u john'. Files owned by deleted user become orphaned (no owner) – find them with 'find / -nouser -ls'. Consider archiving data before deletion.

Q34 What is the /etc/passwd file structure?

Answer Each line has 7 colon-separated fields: username:password:UID:GID:GECOS:home_dir:shell. Example: 'john:x:1001:1001:John Doe:/home/john:/bin/bash'. The 'x' in password field means password hash is in /etc/shadow. UID 0 = root. UIDs 1-999 = system accounts. UIDs 1000+ = regular users.

Q35 What is the /etc/shadow file and why is it important?

Answer /etc/shadow stores password hashes and aging info, readable only by root (mode 640 or 000). Fields: username:hash:last_changed:min_age:max_age:warn_days:inactive_days:expire_date:reserved. Hash starts with \$6\$ (SHA-512), \$5\$ (SHA-256), or \$2b\$ (bcrypt). '!' prefix means account is locked.

Q36 How do you manage password aging policies?

Answer 'chage -l john' lists current password aging settings. 'chage -M 90 john' sets max password age to 90 days. 'chage -m 7 john' sets minimum password age (7 days). 'chage -W 14 john' warns 14 days before expiry. 'chage -d 0 john' forces password change at next login. 'chage -E 2025-12-31 john' sets account expiry.

Q37 How do you create and manage groups?

Answer 'groupadd groupname' creates a new group. 'groupadd -g 1500 developers' specifies :
GID. 'groupmod -n newname oldname' renames a group. 'groupdel groupname' deletes a group (ensure no users have it as primary group first). 'groups username' shows a user's groups. 'id username' shows UID, GID, and all group memberships.

Q38 What is the difference between primary and supplementary groups?

Answer Every user has one primary group (stored in /etc/passwd as GID). New files created by :
that user get this group ownership. Supplementary (secondary) groups provide additional access. 'id' shows all groups. Changing group with 'newgrp groupname' switches active group for the session to affect new file creation.

Q39 How do you grant sudo privileges to a user?

Answer Add user to the sudo group: 'usermod -aG sudo username' (Debian/Ubuntu) or :
'usermod -aG wheel username' (RHEL/CentOS). For specific command access, use 'visudo' to edit /etc/sudoers: 'john ALL=(ALL) /usr/bin/systemctl restart nginx' grants john restart permission only. Never edit /etc/sudoers directly — always use visudo.

Q40 What is the /etc/sudoers file format?

Answer Format: 'user host=(runas_user:runas_group) commands'. 'ALL=(ALL:ALL) ALL' means :
full sudo. '%wheel ALL=(ALL) NOPASSWD: ALL' allows wheel group without password. 'Cmnd_Alias REBOOT = /sbin/reboot, /sbin/shutdown' defines command aliases. Include files with '@includedir /etc/sudoers.d'. Always validate with 'visudo -c'.

Q41 How do you switch between users?

Answer 'su - username' switches to username with a full login shell (loads their environment). :
'su username' switches without loading their environment. 'su -' or 'su - root' switches to root. 'sudo -u username command' runs a single command as another user. 'sudo -i' opens a root interactive login shell.

Q42 What are UID and GID ranges used for?

Answer UID/GID 0: root. 1-99: static system accounts (reserved for distro). 100-999 (or :
500-999 on RHEL): dynamically allocated system accounts for daemons/services. 1000+: regular users. These ranges are defined in /etc/login.defs (UID_MIN, UID_MAX, SYS_UID_MIN, SYS_UID_MAX). 'useradd -r' creates system accounts below UID_MIN.

Q43 How do you configure login.defs?

Answer /etc/login.defs controls default settings for user/group creation. Key directives:

- : PASS_MAX_DAYS (max password age), PASS_MIN_DAYS (min password age), PASS_WARN_AGE (warning period), UID_MIN/UID_MAX (UID range for regular users), ENCRYPT_METHOD (password hashing algorithm like SHA512), CREATE_HOME (default home creation).

Q44 What is the useradd defaults file?

Answer /etc/default/useradd stores default values used by useradd when options aren't specified. Key settings: GROUP (default primary group), HOME (base home directory path), INACTIVE (days after password expiry to disable account), EXPIRE (account expiry date), SHELL (default shell), SKEL (skeleton directory). View with 'useradd -D'.

Q45 What is the /etc/skel directory?

Answer /etc/skel (skeleton) contains template files copied to a new user's home directory when created with 'useradd -m'. Typically includes .bash_profile, .bashrc, and .bash_logout. Administrators add custom files here (like a welcome README or standard config) to provide consistent default environments for all new users.

Q46 How do you lock and unlock user accounts?

Answer 'passwd -l username' locks the account by prepending '!' to the password hash in /etc/shadow. 'passwd -u username' unlocks it. 'usermod -L username' also locks; 'usermod -U username' unlocks. 'chage -E 0 username' sets expiry to epoch (effectively locks). Check lock status with 'passwd -S username'.

Q47 What is PAM (Pluggable Authentication Modules)?

Answer PAM is a flexible authentication framework that separates authentication logic from applications. Config files in /etc/pam.d/ define module stacks for each service. Four management groups: auth (verify identity), account (access control), session (setup/teardown), password (update credentials). Modules: pam_unix.so, pam_pwquality.so, pam_limits.so.

Q48 How do you enforce password complexity with PAM?

Answer The pam_pwquality module enforces complexity rules. In /etc/pam.d/common-password or /etc/pam.d/system-auth: 'password requisite pam_pwquality.so retry=3 minlen=12 ucredit=-1 lcredit=-1 dcredit=-1 ocredit=-1'. Also configure /etc/security/pwquality.conf. 'minlen=12' sets minimum length; credit values require character types.

Q49 What is the getent command?

Answer 'getent' retrieves entries from system databases (like passwd, group, hosts, shadow).
:
'getent passwd username' shows user info from /etc/passwd or LDAP/NIS. 'getent group groupname' shows group membership. 'getent hosts hostname' resolves a hostname using all configured sources in nsswitch.conf. Useful for testing name resolution.

Q50 How do you find all users with UID 0 (root privilege)?

Answer 'awk -F: '(\$3 == 0) {print}' /etc/passwd' finds all accounts with UID 0. There should be only one: root. Multiple UID 0 accounts are a critical security risk. Also check 'awk -F: '(\$2 != "x") {print}' /etc/passwd' to find accounts with password hashes directly in passwd (not shadow).

Q51 What is the w and who command used for?

Answer 'who' shows currently logged-in users, their terminal, login time, and origin. 'w' provides more detail: idle time, CPU usage (JCPU/PCPU), and current command. 'who -b' shows last system boot time. 'who -r' shows current runlevel. 'users' lists just logged-in usernames. 'last' shows login history from /var/log/wtmp.

Q52 How do you monitor user activity?

Answer 'last' reads /var/log/wtmp for login history. 'lastb' reads /var/log/btmp for failed logins. 'lastlog' shows last login for every user. 'audit' framework (auditd) provides detailed tracking. 'history' shows command history (per user, in ~/.bash_history). 'acct' package provides process accounting with 'sa' and 'ac' commands.

Q53 What is the newgrp command?

Answer 'newgrp groupname' switches the user's active primary group to another group they belong to, affecting file creation ownership in that session. It starts a new shell. 'newgrp -' resets to original primary group. Useful when you need files created under a different group without logging out.

Q54 How do you set file ownership?

Answer 'chown user:group file' changes both owner and group. 'chown user file' changes only owner. 'chown :group file' changes only group. 'chgrp group file' is an alternative for group-only changes. 'chown -R user:group directory/' applies recursively. Only root can change file owner; users can change group to one they belong to.

Q55 What is ACL and how do you use it?

Answer ACLs (Access Control Lists) provide fine-grained permissions beyond owner/group/other. 'setfacl -m u:alice:rw file' grants alice read-write. 'setfacl -m g:devs:r file' grants devs group read. 'getfacl file' displays ACL. 'setfacl -x u:alice file' removes alice's ACL. 'setfacl -b file' removes all ACLs. Requires filesystem ACL support (ext4, xfs).

Q56 What is the umask and how does it work?

Answer umask subtracts permissions from the maximum when creating files. Default max: 666 for files, 777 for directories. 'umask 022' results in 644 files (rw-r--r--) and 755 directories (rwxr-xr-x). 'umask' shows current value. Set in ~/.bashrc for user-level defaults. System default often in /etc/profile or /etc/login.defs.

Q57 How do you audit user account changes?

Answer The auditd system tracks changes. 'auditctl -w /etc/passwd -p wa -k user_changes' watches passwd file. 'auditctl -w /etc/shadow -p wa -k shadow_changes' tracks shadow. 'ausearch -k user_changes' searches audit log. Review /var/log/secure or /var/log/auth.log for authentication events. 'journalctl _COMM=useradd' finds useradd events.

Q58 What is the faillock command?

Answer faillock (from pam_faillock) tracks and manages failed login attempts to enforce account lockout. 'faillock' shows current failed attempts for all users. 'faillock --reset --user john' resets john's failed count. Configure in /etc/security/faillock.conf: deny=5 (lock after 5 failures), unlock_time=600 (unlock after 10 minutes).

Q59 How do you configure account lockout policy?

Answer Use pam_faillock in /etc/pam.d/system-auth (RHEL) or /etc/pam.d/common-auth (Debian). Add: 'auth required pam_faillock.so preauth silent deny=5 unlock_time=900'. Configure /etc/security/faillock.conf for centralized settings. Monitor with 'faillock --user username'. Unlock with 'faillock --reset --user username'.

Q60 What is the difference between useradd and adduser?

Answer 'useradd' is the low-level utility — it creates the account but requires all options explicitly (may not create home dir by default on all distros). 'adduser' (Debian/Ubuntu) is a higher-level, interactive Perl script that wraps useradd — it prompts for password, full name, creates home, and copies skel automatically. Use adduser for interactive creation on Debian.

Section 3: Storage Management

Questions 61 – 90

Q61 How do you list block devices and disk layout?

Answer 'lsblk' displays all block devices in a tree format showing name, size, type, and mount point. 'lsblk -f' also shows filesystem type and UUID. 'fdisk -l' lists partition tables. 'blkid' shows device UUIDs and filesystem types. 'ls -la /dev/disk/by-id/' shows persistent device naming by hardware ID.

Q62 What is the difference between MBR and GPT?

Answer MBR (Master Boot Record): legacy partitioning, max 4 primary partitions (3 + 1 extended), max disk size 2TB, 32-bit partition entries. GPT (GUID Partition Table): modern standard, up to 128 partitions, supports disks >2TB, 64-bit LBA addresses, CRC32 checksums for data integrity, requires UEFI (though can boot with BIOS+GPT with workarounds).

Q63 How do you partition a disk with fdisk?

Answer 'fdisk /dev/sdb' opens the interactive partitioning tool for MBR disks. Commands: 'p' (print partition table), 'n' (new partition), 'd' (delete), 't' (change type), 'w' (write and exit), 'q' (quit without saving). After creating partitions, format with 'mkfs.ext4 /dev/sdb1' and add to /etc/fstab.

Q64 How do you partition a disk with parted or gdisk?

Answer 'parted /dev/sdb' supports both MBR and GPT. Commands: 'mklabel gpt', 'mkpart primary ext4 1MiB 50GiB', 'print', 'quit'. 'gdisk /dev/sdb' is specifically for GPT (like fdisk but for GPT). 'parted -s /dev/sdb mklabel gpt mkpart primary ext4 1MiB 100%' is a non-interactive one-liner.

Q65 How do you format a partition with a filesystem?

Answer 'mkfs.ext4 /dev/sdb1' formats with ext4. 'mkfs.xfs /dev/sdb1' formats with XFS. 'mkfs.vfat /dev/sdb1' formats with FAT32. 'mkfs -t ext4 /dev/sdb1' is equivalent. Options: 'mkfs.ext4 -L mydata /dev/sdb1' sets a volume label. Always confirm the correct device before formatting — data will be destroyed.

Q66 How do you mount and unmount filesystems?

Answer `'mount /dev/sdb1 /mnt/data'` mounts the filesystem. `'mount -t ext4 /dev/sdb1 /mnt/data'` specifies type. `'mount -o ro /dev/sdb1 /mnt/data'` mounts read-only. `'umount /mnt/data'` unmounts (also `'umount /dev/sdb1'`). `'umount -l'` lazy unmount if busy. `'lsof /mnt/data'` finds processes using the mount before unmounting.

Q67 What is `/etc/fstab` and what are its fields?

Answer `/etc/fstab` defines persistent mounts. Six fields: (1) device (UUID=xxx or `/dev/sdX` or LABEL=), (2) mount point, (3) filesystem type, (4) options (defaults,ro,noexec), (5) dump (0 or 1), (6) fsck order (0=skip, 1=root, 2=others). `'mount -a'` mounts all fstab entries. `'findmnt --verify'` validates fstab syntax.

Q68 What are common mount options?

Answer defaults (rw,suid,dev,exec,auto,nouser,async). ro/rw (read-only/read-write). noexec (no execution of binaries). nosuid (ignore suid/sgid bits). noatime (don't update access times — improves performance). relatime (update atime only if older than mtime). user (allow user to mount). nofail (don't fail boot if device missing). x-systemd.automount (automount).

Q69 What is ext4 and what are its features?

Answer ext4 (fourth extended filesystem) is the default on most Linux distros. Features: journaling (protects against corruption on crash), extents (efficient large file handling), backwards compatibility with ext2/ext3, max file size 16TiB, max volume size 1EiB, delayed allocation, online defragmentation support, nanosecond timestamps, and extended attributes.

Q70 What is XFS and when should you use it?

Answer XFS is a high-performance journaling filesystem developed by SGI. Default in RHEL 7+. Strengths: excellent for large files and high-throughput workloads, supports online resize (grow only — can't shrink), parallel I/O, up to 8 exabytes, delayed logging. Use for database servers, media storage, NFS exports. `'xfs_repair'` fixes XFS; `'xfs_fsr'` defragments.

Q71 What is Btrfs and its key features?

Answer Btrfs (B-tree filesystem) is a modern copy-on-write (CoW) filesystem. Features: snapshots (instant, space-efficient), subvolumes, built-in RAID (0,1,5,6,10), online resize (grow and shrink), checksums on data and metadata, transparent compression (zstd, lzo, zlib), send/receive for incremental backups. `'btrfs subvolume list /'` lists subvolumes.

Q72 What is LVM and what are its components?

Answer LVM (Logical Volume Manager) creates an abstraction over physical storage. Three layers: Physical Volumes (PV) — actual disks or partitions initialized with 'pvcreate'; Volume Groups (VG) — pool of PVs created with 'vgcreate'; Logical Volumes (LV) — virtual partitions from VG pool created with 'lvcreate'. Allows resizing, snapshots, and spanning disks.

Q73 How do you set up LVM from scratch?

Answer 1) 'pvcreate /dev/sdb /dev/sdc' initializes PVs. 2) 'vgcreate datavg /dev/sdb /dev/sdc' creates volume group. 3) 'lvcreate -L 50G -n datalv datavg' creates 50GB LV. 4) 'mkfs.ext4 /dev/datavg/datalv' formats it. 5) Add to /etc/fstab: '/dev/datavg/datalv /mnt/data ext4 defaults 0 2'. 6) 'mount -a' to mount.

Q74 How do you extend an LVM logical volume?

Answer 'lvextend -L +10G /dev/datavg/datalv' adds 10GB to the LV. 'lvextend -l +100%FREE /dev/datavg/datalv' uses all free space. Then resize the filesystem: 'resize2fs /dev/datavg/datalv' for ext4 (online), or 'xfs_growfs /mnt/data' for XFS. 'lvresize -r -L +10G /dev/datavg/datalv' does both steps at once (-r flag).

Q75 How do you create an LVM snapshot?

Answer 'lvcreate -s -n snap01 -L 5G /dev/datavg/datalv' creates a 5GB snapshot of datalv. Snapshots capture state at creation time using CoW. Mount read-only: 'mount -o ro /dev/datavg/snap01 /mnt/snap'. Restore: 'lvconvert --merge /dev/datavg/snap01' (requires unmounting). Remove with 'lvremove /dev/datavg/snap01'.

Q76 How do you reduce an LVM logical volume?

Answer Reducing LV is risky — must reduce filesystem first, then LV. For ext4: 1) Unmount. 2) 'e2fsck -f /dev/datavg/datalv'. 3) 'resize2fs /dev/datavg/datalv 40G'. 4) 'lvreduce -L 40G /dev/datavg/datalv'. XFS cannot be shrunk. Always backup before reducing. 'lvresize -r -L 40G /dev/datavg/datalv' attempts to do it safely.

Q77 What is RAID and what are the common RAID levels?

Answer RAID combines multiple disks for performance or redundancy. RAID 0: striping — speed, no redundancy, min 2 disks. RAID 1: mirroring — full redundancy, min 2 disks. RAID 5: distributed parity — balanced, can lose 1 disk, min 3 disks. RAID 6: dual parity — can lose 2 disks, min 4 disks. RAID 10: mirrored stripes — speed + redundancy, min 4 disks.

Q78 How do you create a software RAID with mdadm?

Answer 'mdadm --create /dev/md0 --level=1 --raid-devices=2 /dev/sdb /dev/sdc' creates RAID 1. Monitor: 'cat /proc/mdstat'. Save config: 'mdadm --detail --scan >> /etc/mdadm/mdadm.conf'. Format: 'mkfs.ext4 /dev/md0'. Check status: 'mdadm --detail /dev/md0'. Add spare: 'mdadm /dev/md0 --add /dev/sdd'.

Q79 What is swap and how do you create it?

Answer Swap is disk space used as virtual memory overflow when RAM is full. Create swap partition: 'mkswap /dev/sdb2' then 'swapon /dev/sdb2'. Create swap file: 'dd if=/dev/zero of=/swapfile bs=1M count=2048' (2GB), 'chmod 600 /swapfile', 'mkswap /swapfile', 'swapon /swapfile'. Add to /etc/fstab: '/swapfile none swap sw 0 0'. 'swapon -s' shows active swap.

Q80 How do you check and repair filesystems?

Answer 'fsck /dev/sdb1' checks and repairs (must be unmounted first). 'fsck -y /dev/sdb1' auto-answers yes to all repairs. 'e2fsck -f /dev/sdb1' forces check on ext4. 'xfs_repair /dev/sdb1' repairs XFS (must be unmounted). For root filesystem: boot to rescue mode or live media to run fsck. fsck runs automatically at boot on dirty filesystems.

Q81 What is the du command and how do you use it?

Answer 'du -sh /path' shows human-readable total size of a directory. 'du -ah /path | sort -rh | head -20' lists 20 largest files/dirs. 'du --max-depth=1 -h /var' shows first-level subdirectory sizes. 'du -sh /*' shows top-level directory sizes. Combine with 'ncdu' for an interactive TUI disk usage analyzer.

Q82 What is the df command?

Answer 'df -h' shows disk space usage of all mounted filesystems in human-readable format. Columns: Filesystem, Size, Used, Available, Use%, Mounted on. 'df -i' shows inode usage (important when disk has free space but 'no space left' errors occur due to inode exhaustion). 'df -T' also shows filesystem type.

Q83 How do you manage disk quotas?

Answer 1) Add 'usrquota,grpquota' to fstab mount options. 2) Remount: 'mount -o remount /home'. 3) 'quotacheck -cum /home' creates quota database. 4) 'quotaon /home' enables quotas. 5) 'edquota -u username' edits soft/hard limits for blocks and inodes. 6) 'repquota /home' shows quota report. 'quota -u username' shows user's usage.

Q84 What is the purpose of the tune2fs command?

Answer 'tune2fs' adjusts ext2/3/4 filesystem parameters. 'tune2fs -l /dev/sdb1' lists filesystem info. 'tune2fs -L newlabel /dev/sdb1' sets volume label. 'tune2fs -m 1 /dev/sdb1' reduces reserved blocks from 5% to 1% (reclaims space on large data volumes). 'tune2fs -i 0 /dev/sdb1' disables time-based fsck checks.

Q85 How do you identify the UUID of a filesystem?

Answer 'blkid /dev/sdb1' shows UUID, filesystem type, and label. 'blkid' without arguments lists all block devices. 'lsblk -f' also shows UUIDs. 'tune2fs -l /dev/sdb1 | grep UUID' for ext filesystems. UUIDs are stable across reboots and device renaming — always use UUID= in /etc/fstab instead of /dev/sdX device names.

Q86 What is iSCSI and how is it used in Linux?

Answer iSCSI (Internet Small Computer Systems Interface) provides block-level storage over TCP/IP networks. The target (server) exports storage; the initiator (client) connects to it. 'iscsiadm -m discovery -t sendtargets -p target_ip' discovers targets. 'iscsiadm -m node -T targetname -p ip --login' connects. iSCSI block devices appear as /dev/sd* and can be partitioned/formatted normally.

Q87 How do you persistently configure storage with systemd mount units?

Answer Create /etc/systemd/system/mnt-data.mount: [Unit] Description=Data Mount, [Mount] What=/dev/sdb1 Where=/mnt/data Type=ext4 Options=defaults, [Install] WantedBy=multi-user.target. Then 'systemctl daemon-reload && systemctl enable --now mnt-data.mount'. Automount units (.automount) mount only when accessed, saving resources.

Q88 What is the difference between dd and cp for copying disks?

Answer 'dd' copies block-by-block, regardless of filesystem, making exact bit-for-bit copies including empty space. 'dd if=/dev/sda of=/dev/sdb bs=4M status=progress' clones a disk. 'cp -a' copies at filesystem level — only copies actual files, preserving attributes. dd is used for disk imaging; cp is used for file/directory copying. dd can be destructive — double-check if/of arguments.

Q89 What is NFS and how do you mount an NFS share?

Answer NFS (Network File System) shares directories over a network. Server: add to /etc/exports: '/shared 192.168.1.0/24(rw,sync,no_subtree_check)', then 'exportfs -ra' and start nfs-server. Client: 'mount -t nfs server:/shared /mnt/nfs' or in /etc/fstab: 'server:/shared /mnt/nfs nfs defaults,_netdev 0 0'. 'showmount -e server' lists exports.

Q90 What is the autofs service?

Answer : autofs automatically mounts filesystems (NFS, Samba, local) when accessed and unmounts after an idle timeout. Main config: /etc/auto.master defines mount points and map files. /etc/auto.misc defines actual mounts. Example in auto.misc: 'data -fstype=nfs,rw server:/data'. autofs saves resources by not keeping unused mounts active.

Section 4: Networking

Questions 91 – 120

Q91 How do you configure a static IP address in Linux?

Answer Using NetworkManager: 'nmcli connection modify eth0 ipv4.method manual : ipv4.addresses 192.168.1.10/24 ipv4.gateway 192.168.1.1 ipv4.dns 8.8.8.8', then 'nmcli connection up eth0'. Debian/Ubuntu without NM: edit /etc/network/interfaces. RHEL without NM: edit /etc/sysconfig/network-scripts/ifcfg-eth0. Always specify BOOTPROTO=none for static.

Q92 How do you configure DNS resolution?

Answer /etc/resolv.conf specifies nameservers: 'nameserver 8.8.8.8', 'nameserver 1.1.1.1', and : 'search example.com'. /etc/nsswitch.conf defines lookup order: 'hosts: files dns myhostname'. /etc/hosts provides local overrides. On systemd systems, systemd-resolved manages DNS — 'resolvectl status' shows resolver info. 'dig @8.8.8.8 domain.com' tests specific DNS servers.

Q93 What is the ip command and its common uses?

Answer 'ip addr show' (or 'ip a'): show IP addresses. 'ip link show': show interfaces. 'ip route : show' (or 'ip r'): routing table. 'ip addr add 192.168.1.10/24 dev eth0': assign IP. 'ip link set eth0 up/down': enable/disable interface. 'ip route add default via 192.168.1.1': add default gateway. 'ip neigh show': ARP table. Changes are not persistent — use NetworkManager or config files.

Q94 What is NetworkManager and how do you use nmcli?

Answer NetworkManager manages network connections dynamically. nmcli is its CLI. 'nmcli : device status': show all devices. 'nmcli connection show': list connections. 'nmcli connection up/down NAME': activate/deactivate. 'nmcli device wifi list': scan WiFi. 'nmcli device wifi connect SSID password PASS': connect to WiFi. 'nmcli general hostname': show/set hostname.

Q95 How do you troubleshoot network connectivity?

Answer Systematic approach: 1) 'ip link show' — is interface up? 2) 'ip addr show' — do we have : an IP? 3) 'ping 127.0.0.1' — loopback OK? 4) 'ping gateway_ip' — can reach local network? 5) 'ping 8.8.8.8' — internet reachable? 6) 'ping google.com' — DNS working? 7) 'traceroute google.com' — where does it fail? 8) 'ss -tuln' — are expected services listening?

Q96 What is the ss command?

Answer 'ss' (socket statistics) replaces the deprecated netstat. 'ss -tln': all listening TCP/UDP sockets with port numbers (no DNS lookup). 'ss -tp': TCP connections with process names. 'ss -s': socket summary. 'ss -4 state established': established IPv4 connections. 'ss -ltnp | grep :22': check if SSH is listening. Much faster than netstat on systems with many connections.

Q97 How do you configure firewall rules with firewalld?

Answer firewalld uses zones and services. 'firewall-cmd --state': check status. 'firewall-cmd --get-default-zone': current zone. 'firewall-cmd --zone=public --add-service=http --permanent': allow HTTP permanently. 'firewall-cmd --zone=public --add-port=8080/tcp --permanent': allow port 8080. 'firewall-cmd --reload': apply changes. 'firewall-cmd --list-all': show zone rules.

Q98 How do you configure firewall rules with ufw?

Answer ufw (Uncomplicated Firewall) is the default on Ubuntu. 'ufw enable': activate. 'ufw status verbose': show rules. 'ufw allow ssh': allow SSH. 'ufw allow 80/tcp': allow HTTP. 'ufw deny from 192.168.1.100': block IP. 'ufw allow from 192.168.1.0/24 to any port 22': restrict SSH to subnet. 'ufw delete allow 80/tcp': remove rule. Rules are persistent.

Q99 What is iptables and how do you use it?

Answer iptables uses tables (filter, nat, mangle) with chains (INPUT, OUTPUT, FORWARD). 'iptables -L -n -v': list all rules. 'iptables -A INPUT -p tcp --dport 22 -j ACCEPT': allow SSH. 'iptables -A INPUT -j DROP': drop all other input. 'iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT': allow established connections. Save rules: 'iptables-save > /etc/iptables/rules.v4'.

Q100 What is the difference between firewalld, ufw, and iptables?

Answer iptables: low-level, stateless, requires manual rule ordering, not persistent by default. nftables: modern iptables replacement, cleaner syntax, same level. ufw: simple frontend for iptables on Ubuntu/Debian — good for basic firewalling. firewalld: zone-based frontend used on RHEL/CentOS, supports dynamic rule changes without restarting, uses nftables or iptables backend.

Q101 How do you configure a network bridge?

Answer A bridge connects network segments at layer 2. Create: 'ip link add br0 type bridge', 'ip link set eth0 master br0', 'ip link set br0 up'. With NetworkManager: 'nmcli connection add type bridge ifname br0'. Bridges are used for virtual machine networking (KVM) and container networking. 'brctl show' lists bridges (legacy command).

Q102 What is network bonding/teaming?

Answer Bonding combines multiple physical interfaces for redundancy or throughput. 'ip link add bond0 type bond', 'ip link set bond0 type bond mode active-backup', 'ip link set eth0 master bond0', 'ip link set eth1 master bond0'. Modes: active-backup (failover), balance-rr (round-robin), 802.3ad (LACP). NetworkManager: 'nmcli connection add type bond'. Check: 'cat /proc/net/bonding/bond0'.

Q103 What is a VLAN and how do you configure it?

Answer VLANs (Virtual LANs) segment a physical network into logical networks. Create VLAN interface: 'ip link add link eth0 name eth0.100 type vlan id 100', 'ip link set eth0.100 up', 'ip addr add 192.168.100.1/24 dev eth0.100'. With nmcli: 'nmcli connection add type vlan con-name vlan100 dev eth0 id 100'. 'cat /proc/net/vlan/eth0.100' shows VLAN config.

Q104 How do you set up SSH port forwarding?

Answer Local forwarding: 'ssh -L 8080:internal-server:80 user@bastion' — connects localhost:8080 to internal-server:80 through bastion. Remote forwarding: 'ssh -R 9090:localhost:3000 user@server' — exposes local port 3000 on server:9090. Dynamic forwarding (SOCKS proxy): 'ssh -D 1080 user@server'. Add -N (no command) and -f (background) for clean forwarding sessions.

Q105 What is the netplan configuration system?

Answer Netplan is the default network configuration tool on Ubuntu 18.04+. Config files are YAML in /etc/netplan/. Example static IP config: 'network: version: 2, renderer: networkd, ethernets: eth0: addresses: [192.168.1.10/24], gateway4: 192.168.1.1, nameservers: addresses: [8.8.8.8]'. Apply with 'netplan apply'. Test first with 'netplan try' (auto-reverts after 120s if not confirmed).

Q106 How do you capture and analyze network traffic?

Answer 'tcpdump -i eth0' captures all traffic on eth0. 'tcpdump -i eth0 port 80' filters HTTP. 'tcpdump -i eth0 -w capture.pcap' saves to file. 'tcpdump -r capture.pcap' reads saved file. Wireshark provides GUI analysis of pcap files. 'tcpdump -nn -X port 443' shows raw hex/ASCII. Requires root or CAP_NET_ADMIN capability.

Q107 What is the difference between TCP and UDP?

Answer TCP (Transmission Control Protocol): connection-oriented, reliable delivery, ordered, error correction, slower — used for HTTP, HTTPS, SSH, FTP, SMTP. UDP (User Datagram Protocol): connectionless, no guaranteed delivery, no ordering, faster — used for DNS, DHCP, NTP, streaming, VoIP. TCP has handshake and acknowledgments; UDP is fire-and-forget. 'ss -tu' shows both types.

Q108 How do you test if a port is open on a remote host?

Answer 'nc -zv host port' (netcat) tests connectivity (z=scan mode, v=verbose). 'telnet host port' is another classic test. 'nmap -p port host' scans specific port. 'curl -v telnet://host:port' works for TCP. In scripts, use `bash: timeout 3 bash -c '/dev/tcp/host/port redirect'` and check exit status. 'ss' and 'netstat' check local ports only.

Q109 What is name resolution and how does it work in Linux?

Answer Name resolution converts hostnames to IPs. Process: 1) Check `/etc/nsswitch.conf` for 'hosts' order (typically 'files dns'). 2) Check `/etc/hosts` for static entries. 3) Query DNS servers in `/etc/resolv.conf`. 4) systemd-resolved intercepts DNS if active. 'resolvectl query hostname' shows resolution details. 'getent hosts hostname' tests resolution using nsswitch.conf chain.

Q110 What is the curl command and how is it used?

Answer 'curl URL' fetches content. 'curl -o file.zip URL' saves to file. 'curl -I URL' fetches headers only. 'curl -X POST -d '{"key":"val"}' -H 'Content-Type: application/json' URL' sends JSON POST. 'curl -u user:pass URL' HTTP auth. 'curl -k URL' skips SSL verification. 'curl --proxy http://proxy:3128 URL' uses proxy. 'curl -v URL' verbose output for debugging.

Q111 How do you configure a static route?

Answer Temporary: 'ip route add 10.10.0.0/16 via 192.168.1.254 dev eth0'. Delete: 'ip route del 10.10.0.0/16'. Persistent with NetworkManager: 'nmcli connection modify eth0 +ipv4.routes "10.10.0.0/16 192.168.1.254"'. On RHEL: add to `/etc/sysconfig/network-scripts/route-eth0`: '10.10.0.0/16 via 192.168.1.254'. 'ip route show' verifies current routes.

Q112 What is IPv6 and how do you configure it in Linux?

Answer IPv6 uses 128-bit addresses (e.g., 2001:db8::1/64). 'ip -6 addr show': show IPv6 addresses. 'ip -6 route show': IPv6 routing table. 'ping6 ::1': test IPv6 loopback. fe80:: prefix: link-local (auto-configured). SLAAC (Stateless Address Autoconfiguration) auto-assigns addresses from router advertisements. Disable IPv6: 'sysctl -w net.ipv6.conf.all.disable_ipv6=1'.

Q113 What is the purpose of `/etc/hosts`?

Answer `/etc/hosts` provides static hostname-to-IP mappings, checked before DNS (if configured so in `nsswitch.conf`). Format: 'IP hostname aliases'. Always contains '127.0.0.1 localhost' and '::1 localhost'. Used for: development environments, blocking sites (point to 127.0.0.1), container networking, overriding DNS. Changes take effect immediately without restart.

Q114 How do you monitor bandwidth usage?

Answer 'iftop -i eth0' shows real-time per-connection bandwidth usage. 'nload eth0' shows interface throughput graphs. 'nethogs' shows bandwidth per process. 'vnstat' collects historical bandwidth statistics. 'ip -s link show eth0' shows raw packet/byte counters. 'cat /proc/net/dev' shows interface statistics. 'sar -n DEV 1' from sysstat shows periodic interface stats.

Q115 What is a proxy server and how do you configure Linux to use one?

Answer A proxy routes traffic through an intermediary. Set system-wide: `export http_proxy='http://proxy:3128', https_proxy='http://proxy:3128', no_proxy='localhost,127.0.0.1'`. Add to `/etc/environment` for persistence. apt uses 'Acquire::http::Proxy' in `/etc/apt/apt.conf.d/proxy.conf`. curl/wget respect http_proxy. GNOME/KDE have GUI proxy settings.

Q116 What is multipathing in Linux?

Answer Multipathing (DM-Multipath) provides redundant paths to storage devices (typically SAN). If one path fails, traffic automatically uses another. Config: `/etc/multipath.conf`. 'multipathd' daemon manages paths. 'multipath -ll' shows multipath topology. 'mpathconf --enable' sets up default config on RHEL. Devices appear as `/dev/mapper/mpathX`.

Q117 How do you configure NTP with chrony?

Answer Install chrony. Edit `/etc/chrony.conf`: add 'server pool.ntp.org iburst'. 'systemctl enable --now chronyd'. Verify: 'chronyc sources -v' shows time sources. 'chronyc tracking' shows sync status. 'timedatectl show' shows NTP status. 'chronyc makestep' forces immediate time adjustment. For servers in secure networks, configure an internal NTP server as source.

Q118 What is the purpose of the `/etc/services` file?

Answer `/etc/services` maps service names to port numbers and protocols. Format: 'ssh 22/tcp'. Used by tools like netstat, ss, and network applications to resolve port names to numbers. Does not control which services run — it's a reference database. 'getent services ssh' looks up a service. Some tools like nmap also use this file for service identification.

Q119 What is the difference between ssh and scp?

Answer : SSH provides encrypted remote shell access: 'ssh user@host'. SCP copies files securely over SSH: 'scp file user@host:/path' (local to remote), 'scp user@host:/path/file .' (remote to local), 'scp -r dir/ user@host:/path' (recursive). SFTP is an alternative: 'sftp user@host'. rsync over SSH is more efficient for large transfers: 'rsync -avz -e ssh src/ user@host:/dst/'.

Q120 What is the wget command?

Answer : 'wget URL' downloads a file to current directory. 'wget -O filename URL' saves with a specific name. 'wget -r -l2 URL' recursively downloads up to 2 levels deep. 'wget -c URL' resumes an interrupted download. 'wget -q URL' runs quietly. 'wget --limit-rate=100k URL' limits download speed. 'wget -i urls.txt' downloads from a list of URLs. Useful for scripted/unattended downloads.

Section 5: Service Management & Security

Questions 121 – 150

Q121 How do you configure SSH for key-based authentication?

Answer 1) Generate key pair: 'ssh-keygen -t ed25519 -C "email@example.com"'. 2) Copy public key to server: 'ssh-copy-id user@server' or manually append ~/.ssh/id_ed25519.pub to ~/.ssh/authorized_keys on server. 3) Set permissions: 'chmod 700 ~/.ssh && chmod 600 ~/.ssh/authorized_keys'. 4) Test: 'ssh -i ~/.ssh/id_ed25519 user@server'. 5) Disable password auth in /etc/ssh/sshd_config: 'PasswordAuthentication no'.

Q122 What are important SSH server security settings?

Answer In /etc/ssh/sshd_config: 'PermitRootLogin no' (disable root login). 'PasswordAuthentication no' (keys only). 'PubkeyAuthentication yes'. 'AllowUsers alice bob' (whitelist users). 'MaxAuthTries 3' (limit attempts). 'ClientAliveInterval 300' (disconnect idle clients). 'Port 2222' (non-standard port). 'Protocol 2' (SSH2 only). Restart: 'systemctl restart sshd'. Validate: 'sshd -t'.

Q123 What is SELinux and how do you manage it?

Answer SELinux (Security-Enhanced Linux) provides Mandatory Access Control. Modes: Enforcing (blocks and logs), Permissive (logs only), Disabled. 'getenforce' shows current mode. 'setenforce 0' temporarily sets permissive. Permanent: edit /etc/selinux/config, set SELINUX=enforcing. 'sestatus' shows detailed status. 'ausearch -m AVC' finds denials. 'semanage' and 'chcon' manage file contexts.

Q124 How do you troubleshoot SELinux denials?

Answer 'ausearch -m AVC -ts recent' shows recent denials. 'sealert -a /var/log/audit/audit.log' provides human-readable analysis with suggested fixes. 'audit2allow -w -a' explains denials. 'audit2allow -a -M mypolicy && semodule -i mypolicy.pp' creates and installs a policy module to allow denied actions. 'restorecon -Rv /path' restores correct file contexts.

Q125 What is AppArmor and how do you manage it?

Answer AppArmor confines programs using path-based profiles. 'aa-status' shows loaded profiles and modes. Modes: enforce, complain (log only), unconfined. 'aa-enforce /etc/apparmor.d/usr.sbin.nginx' puts profile in enforce mode. 'aa-complain profile' switches to complain. 'apparmor_parser -r /etc/apparmor.d/profile' reloads a profile. Profiles are in /etc/apparmor.d/. 'dmesg | grep apparmor' shows denials.

Q126 How do you configure fail2ban?

Answer fail2ban reads log files and bans IPs after repeated failures. Main config: `/etc/fail2ban/jail.conf`. Override in `/etc/fail2ban/jail.local`. Enable SSH jail: `[sshd] enabled=true maxretry=5 bantime=3600`. 'fail2ban-client status' shows active jails. 'fail2ban-client status sshd' shows banned IPs for SSH. 'fail2ban-client set sshd unbanip 1.2.3.4' unbans an IP. 'systemctl restart fail2ban' applies changes.

Q127 What is sudo and how do you configure it securely?

Answer sudo grants temporary elevated privileges. Configure with 'visudo' (always – validates syntax). Grant specific commands: `'alice ALL=(root) /bin/systemctl restart nginx'`. Use command aliases: `'Cmnd_Alias SERVICES = /bin/systemctl'`. Require password: remove NOPASSWD. Enable logging: `'Defaults logfile=/var/log/sudo.log'`. Limit to specific hosts. Use groups: `'%ops ALL=(ALL) ALL'`. Audit with `'grep sudo /var/log/auth.log'`.

Q128 What is disk encryption and how do you set up LUKS?

Answer LUKS (Linux Unified Key Setup) encrypts block devices. 1) `'cryptsetup luksFormat /dev/sdb1'` initializes encryption. 2) `'cryptsetup open /dev/sdb1 mydata'` opens/decrypts to `/dev/mapper/mydata`. 3) `'mkfs.ext4 /dev/mapper/mydata'` formats. 4) `'mount /dev/mapper/mydata /mnt/data'` mounts. For boot: add to `/etc/crypttab`. `'cryptsetup luksAddKey'` adds additional keys/passphrases. `'luksDump'` shows header info.

Q129 How do you manage SSL/TLS certificates?

Answer `'openssl req -newkey rsa:2048 -keyout key.pem -out csr.pem'` generates key+CSR. `'openssl req -x509 -newkey rsa:2048 -keyout key.pem -out cert.pem -days 365 -nodes'` creates self-signed cert. `'openssl x509 -in cert.pem -text -noout'` reads cert details. `'openssl s_client -connect host:443'` tests SSL. Let's Encrypt: `'certbot --nginx -d example.com'` for free trusted certs.

Q130 What are file permission special bits (SUID, SGID, Sticky)?

Answer SUID (4xxx): executable runs as file owner, not current user (e.g., `/usr/bin/passwd` runs as root). `'chmod u+s file'` or `'chmod 4755 file'`. SGID (2xxx): executable runs with file's group; on directories, new files inherit directory's group. `'chmod g+s'`. Sticky bit (1xxx): on directories, only owner can delete their files (e.g., `/tmp` has 1777). `'chmod +t dir'`. Find SUID: `'find / -perm -4000 -ls'`.

Q131 How do you audit the system for security?

Answer auditd is the primary audit framework. 'auditctl -w /etc/passwd -p wa -k passwd_changes' watches file writes/attribute changes. 'auditctl -a always,exit -F arch=b64 -S execve -k exec_log' logs all executed commands. 'ausearch -k passwd_changes' searches by key. 'aureport -au' shows authentication report. 'aureport -x --summary' shows executed commands summary. Config: /etc/audit/audit.rules.

Q132 What is the auditd daemon?

Answer auditd (Linux Audit Daemon) records security-relevant kernel events to /var/log/audit/audit.log. Components: auditd (daemon), auditctl (rule management), ausearch (log search), aureport (report generation), audispd (dispatcher for plugins). Rules survive reboots when in /etc/audit/rules.d/. Use for PCI-DSS, HIPAA, and SOX compliance. 'systemctl status auditd' checks health.

Q133 How do you configure password quality requirements?

Answer pam_pwquality module enforces complexity. In /etc/security/pwquality.conf: 'minlen=12' (min length), 'minclass=3' (require 3 character classes), 'maxrepeat=3' (max repeated chars), 'dcredit=-1' (require digit), 'ucredit=-1' (require uppercase), 'lcredit=-1' (require lowercase). Also configure PAM stack in /etc/pam.d/common-password to use pam_pwquality.so.

Q134 What is the difference between authentication and authorization?

Answer Authentication (AuthN): verifying identity — 'Who are you?' Proves via password, SSH key, certificate, biometrics, or MFA. Linux uses PAM for authentication. Authorization (AuthZ): determining what an authenticated identity can do — 'What are you allowed to do?' Linux uses file permissions, ACLs, sudo rules, SELinux/AppArmor policies, and capabilities for authorization.

Q135 What are Linux capabilities?

Answer Capabilities break root privilege into distinct units, allowing processes to have specific elevated permissions without full root. Examples: CAP_NET_BIND_SERVICE (bind ports <1024), CAP_NET_ADMIN (network config), CAP_SYS_TIME (set clock), CAP_DAC_OVERRIDE (bypass file permissions). 'getcap /usr/bin/ping' shows capabilities. 'setcap cap_net_bind_service+ep /usr/bin/node' grants capability to a binary.

Q136 How do you harden SSH configuration?

Answer Key hardening steps: 'PermitRootLogin no', 'PasswordAuthentication no', 'PubkeyAuthentication yes', 'Protocol 2', 'MaxAuthTries 3', 'X11Forwarding no', 'AllowTcpForwarding no' (if not needed), 'ClientAliveInterval 300 ClientAliveCountMax 2', 'AllowUsers admin deploy', 'Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com' (strong ciphers only). Test config: 'sshd -t'.

Q137 What is TCP wrappers?

Answer TCP wrappers (libwrap) provide host-based access control for network services. /etc/hosts.allow: permitted hosts. /etc/hosts.deny: denied hosts. Evaluated in order: allow first, then deny. Example hosts.allow: 'sshd: 192.168.1.0/24'. Example hosts.deny: 'ALL: ALL'. Check if a service supports TCP wrappers: 'ldd /usr/sbin/sshd | grep libwrap'. On modern systems, firewall rules are preferred.

Q138 How do you set up a basic web server with Nginx?

Answer 'apt install nginx' or 'yum install nginx'. Config in /etc/nginx/nginx.conf and /etc/nginx/sites-available/. Create site config: 'server { listen 80; server_name example.com; root /var/www/html; index index.html; }'. Enable: 'ln -s /etc/nginx/sites-available/mysite /etc/nginx/sites-enabled/'. Test config: 'nginx -t'. Reload: 'systemctl reload nginx'. Check logs: /var/log/nginx/access.log and error.log.

Q139 How do you configure and manage cron jobs?

Answer 'crontab -e' edits user cron jobs. Format: 'min hour day month weekday command'. '*/*5 * * * * /script.sh' runs every 5 minutes. '0 2 * * 0 /backup.sh' runs at 2 AM Sundays. '0 */6 * * * cmd' runs every 6 hours. System crons: /etc/cron.d/, /etc/cron.daily/, /etc/cron.hourly/. Environment variables can be set at top of crontab. Check cron log: 'grep cron /var/log/syslog'.

Q140 What is a systemd timer and how does it compare to cron?

Answer systemd timers pair a .timer unit with a .service unit. /etc/systemd/system/backup.timer: [Timer] OnCalendar=daily, [Install] WantedBy=timers.target. Advantages over cron: better logging (journalctl -u backup.service), missed runs catch-up (Persistent=true), dependency management, no daemon (activated by systemd). 'systemctl list-timers' shows all timers and next run time.

Q141 What is the rsyslog configuration?

Answer rsyslog receives, processes, and routes log messages. Config: `/etc/rsyslog.conf` and `/etc/rsyslog.d/`. Format: `'facility.severity /path/to/file'`. `*.emerg :omusrmsg:*` sends emergencies to all users. `'auth,authpriv.* /var/log/auth.log'` logs authentication. Remote logging: `*.* @@remote-server:514` (TCP). `'rsyslogd -N1'` validates config. Templates customize log format. `'systemctl restart rsyslog'` applies changes.

Q142 How do you forward logs to a remote syslog server?

Answer In `/etc/rsyslog.conf`: `*.* @remote_ip:514` sends all logs via UDP; `*.* @@remote_ip:514` uses TCP (more reliable). For TLS: use `gtls` driver with certificates. Alternative: `systemd-journal-remote` for journal-native forwarding. Or configure `journal` to forward: add `'ForwardToSyslog=yes'` in `/etc/systemd/journald.conf`. Test: `'logger -p auth.info "test message"'`.

Q143 What is logrotate and how do you configure it?

Answer logrotate manages log rotation, compression, and cleanup. Main config: `/etc/logrotate.conf`. Service configs: `/etc/logrotate.d/`. Example config: `'/var/log/myapp/*.log { daily rotate 14 compress delaycompress missingok notifempty sharedscripts postrotate systemctl reload myapp endscrip }`. `'logrotate -d /etc/logrotate.d/myapp'` dry-run tests config. `'logrotate -f /etc/logrotate.conf'` forces rotation now.

Q144 What is the purpose of `/var/log/auth.log` or `/var/log/secure`?

Answer `/var/log/auth.log` (Debian/Ubuntu) or `/var/log/secure` (RHEL/CentOS) contains authentication events: SSH logins, sudo usage, PAM authentication, su commands, and failed login attempts. Monitor with: `'tail -f /var/log/auth.log'`. Find failed SSH attempts: `'grep 'Failed password' /var/log/auth.log'`. Find sudo usage: `'grep sudo /var/log/auth.log'`. journalctl equivalent: `'journalctl -u sshd'`.

Q145 How do you configure a time server with chrony?

Answer As an NTP server: in `/etc/chrony.conf` add `'allow 192.168.1.0/24'` to permit clients. Also add `'local stratum 10'` to serve time even without upstream sync. Client config: `'server your_ntp_server iburst'`. `'chronyc clients'` shows connected clients. `'chronyc makestep'` forces time sync. `'timedatectl set-ntp true'` enables NTP. Verify: `'chronyc sources -v'` shows source quality.

Q146 What are common security hardening practices for Linux servers?

Answer 1) Disable unused services: 'systemctl disable --now service'. 2) Apply security updates: 'unattended-upgrades' or 'dnf-automatic'. 3) Firewall: allow only necessary ports. 4) SSH hardening (key auth only, no root login). 5) SELinux/AppArmor enforcing. 6) fail2ban for brute force protection. 7) Audit with auditd. 8) Minimal package installation. 9) File integrity monitoring (AIDE/Tripwire). 10) Regular log review.

Q147 How do you manage packages with apt?

Answer 'apt update' refreshes package lists. 'apt upgrade' upgrades installed packages. 'apt install package' installs. 'apt remove package' removes (keeps config). 'apt purge package' removes with config. 'apt autoremove' removes unused dependencies. 'apt search keyword' searches. 'apt show package' shows info. 'apt list --installed' lists installed packages. /etc/apt/sources.list and /etc/apt/sources.list.d/ define repositories.

Q148 How do you manage packages with dnf/yum?

Answer 'dnf install package' installs. 'dnf remove package' removes. 'dnf update' upgrades all. 'dnf search keyword' searches. 'dnf info package' shows details. 'dnf list installed' lists installed. 'dnf history' shows transaction history. 'dnf repolist' lists repos. 'dnf provides /usr/bin/vim' finds which package owns a file. Config: /etc/dnf/dnf.conf. Repos: /etc/yum.repos.d/*.repo.

Q149 What is the difference between rpm and dpkg?

Answer rpm (Red Hat Package Manager) is the low-level package tool for RHEL/CentOS/Fedora. 'rpm -ivh package.rpm' installs. 'rpm -qa' lists all installed. 'rpm -qf /path/to/file' finds package owning a file. 'rpm -V package' verifies. dpkg is the low-level tool for Debian/Ubuntu. 'dpkg -i package.deb' installs. 'dpkg -l' lists installed. 'dpkg -S /path/file' finds owning package. Both are wrapped by dnf/apt.

Q150 How do you compile software from source?

Answer Typical process: 1) Download and extract tarball: 'tar -xzf software.tar.gz'. 2) Enter directory. 3) './configure --prefix=/usr/local' checks dependencies and generates Makefile. 4) 'make -j\$(nproc)' compiles using all CPU cores. 5) 'sudo make install' installs. Uninstall: 'sudo make uninstall' (if supported) or track with 'stow'. Install build essentials first: 'apt install build-essential' or 'dnf groupinstall "Development Tools"'.

Section 6: Shell Scripting & Automation

Questions 151 – 180

Q151 What is a shell script and how do you create one?

Answer A shell script is a text file containing shell commands executed in sequence. Create: 1) Create file with shebang: `'#!/bin/bash'`. 2) Add commands. 3) Make executable: `'chmod +x script.sh'`. 4) Run: `'./script.sh'`. The shebang specifies the interpreter. Use `'#!/usr/bin/env bash'` for portability (finds bash in PATH). Scripts automate repetitive tasks, system administration, and deployments.

Q152 What are positional parameters and special variables in Bash?

Answer `$0`: script name. `$1-$9`: arguments. `${10}+`: use braces for 10+. `$#`: number of arguments. `$@`: all arguments (preserves quoting). `$*`: all arguments as one string. `$?`: exit status of last command (0=success). `$$`: current PID. `$!`: PID of last background command. `$-`: current shell options. `BASH_VERSION`: Bash version. `LINENO`: current line number. `FUNCNAME`: current function name.

Q153 How do you handle errors in shell scripts?

Answer `'set -e'` exits on any error. `'set -u'` exits if undefined variable used. `'set -o pipefail'` catches pipe failures (not just last command). Combine: `'set -euo pipefail'` at top of script. Custom error handling: `'trap 'echo "Error on line $LINENO" && exit 1' ERR'`. Check return codes: `'command || { echo "failed"; exit 1; }'`. Use `'if command; then ... fi'` for controlled error handling.

Q154 What is the difference between `$@` and `$*` in Bash?

Answer Both hold all positional parameters, but behave differently when quoted. `"$@"` expands each argument as a separate quoted string — preserves arguments with spaces: for arg in `"$@"`. `"$*"` expands all arguments as a single string joined by `$IFS` (first char, usually space). Rule: always use `"$@"` when iterating over arguments to correctly handle arguments containing spaces or special characters.

Q155 How do you use arrays in Bash?

Answer Indexed arrays: `'arr=(a b c)'`. Access: `'${arr[0]}'` (first), `'${arr[@]}'` (all), `'${#arr[@]}'` (count), `'${arr[@]:1:2}'` (slice from index 1, 2 elements). Append: `'arr+=(d)'`. Unset: `'unset arr[1]'`. Associative (hash) arrays: `'declare -A hash; hash[key]=value'`. Access: `'${hash[key]}'`. Keys: `'${!hash[@]}'`. Iterate: `'for key in "${!hash[@]}"; do echo "$key=${hash[$key]}"; done'`.

Q156 What is the difference between single and double quotes in Bash?

Answer Single quotes (') preserve every character literally — no variable expansion, no command substitution, no escape sequences interpreted. Double quotes (") allow: \$VAR variable expansion, \$(command) substitution, ` backtick substitution, \n and other backslash escapes, but prevent glob expansion and word splitting. Use single quotes for literal strings; double quotes when you need variable expansion with protection from word splitting.

Q157 How do you perform arithmetic in Bash?

Answer '\$((expr))' performs integer arithmetic: 'echo \$((5 * 3))'. 'let result=5+3'. 'result=\$((a + b))'. 'declare -i n=5; n+=3'. Supports: +, -, *, /, %, ** (power), bit ops (<<, >>, &, |, ^), comparison. For floating point, use bc: 'echo "scale=2; 5/3" | bc' or awk: 'awk "BEGIN{printf \"%.2f\\n\", 5/3}"'. expr is legacy and not recommended.

Q158 How do you use conditional expressions in Bash?

Answer File tests: '[-f file]' (regular file), '[-d dir]' (directory), '[-r file]' (readable), '[-x file]' (executable), '[-s file]' (non-empty). String: '[-z "\$var"]' (empty), '[-n "\$var"]' (non-empty), '["\$a" = "\$b"]' (equal). Integer: '["\$a" -eq "\$b"]' (equal), '-ne, -lt, -le, -gt, -ge'. Use [[]] for Bash-specific features like regex: '[["\$str" =~ ^[0-9]+\$]]'.

Q159 How do you use functions in Bash?

Answer Define: 'function_name() { local var=value; commands; return 0; }'. Call: 'function_name arg1 arg2'. Parameters: \$1, \$2 within function. 'local' restricts variable scope to function. 'return N' sets exit status (0=success). Functions can't return values directly — use echo and command substitution: 'result=\$(get_value)'. Declare before use or use forward declaration pattern.

Q160 What is the here-document (here-doc) syntax?

Answer Here-doc passes multiline input to a command. Basic: 'cat << EOF ... content ... EOF'. Variables are expanded by default. Prevent expansion with quoted delimiter: 'cat << 'EOF' ... no expansion here ... EOF'. Indent here-doc body with <<-EOF (strips leading tabs). 'ssh user@host << EOF commands EOF' runs commands remotely. Useful for config file generation and sending multiline strings.

Q161 How do you debug Bash scripts?

Answer 'bash -x script.sh' runs with debug trace — prints each command before execution. 'set -x' enables tracing from a specific point; 'set +x' disables. 'bash -n script.sh' syntax checks without running. 'set -v' prints source lines as read. Add 'PS4='+\${BASH_SOURCE}:\${LINENO}): ' for rich trace output showing file/line. 'bashdb' provides step-by-step debugging.

Q162 What are string operations in Bash?

Answer Length: `${#var}`. Substring: `${var:2:5}` (from pos 2, length 5). Strip prefix: `${var#prefix}` (shortest), `${var##prefix}` (longest/greedy). Strip suffix: `${var%suffix}`, `${var%%suffix}`. Replace: `${var/pattern/replace}` (first), `${var//pattern/replace}` (all). Upper/lowercase: `${var^^}` (upper), `${var,,}` (lower). Default value: `${var:-default}`. Assign if empty: `${var:=default}`.

Q163 How do you read and process files line by line?

Answer `'while IFS= read -r line; do echo "$line"; done < file.txt'` reads each line safely. `IFS=` preserves leading whitespace. `-r` prevents backslash interpretation. `'while IFS=: read -r user _ uid _; do echo "$user:$uid"; done < /etc/passwd'` reads structured files. Process output of commands: `'command | while IFS= read -r line; do ...; done'`. Avoid `'for line in $(cat file)'` — breaks on spaces.

Q164 How do you use getopt for argument parsing?

Answer `'getopts'` parses command-line options. Example: `'while getopts "f:vh" opt; do case $opt in f) file="$OPTARG";; v) verbose=1;; h) usage; exit;; ?) echo "Invalid option"; exit 1;; esac; done'`. `.'` after option letter means it takes an argument (stored in `$OPTARG`). `$OPTIND` is the index of next argument to process. Shift positional parameters after processing: `'shift $((OPTIND-1))'`.

Q165 What is command substitution in Bash?

Answer Command substitution captures a command's output as a string. Modern syntax: `'$(command)'`. Legacy: ``command`` (avoid — harder to nest). Examples: `'date=$(date +%Y%m%d)'`, `'files=$(ls *.txt)'`, `'count=$(wc -l < file.txt)'`. Nest: `'result=$(echo $(cat file) | sort)'`. Strips trailing newlines. Preserve newlines: `'output=$(cat file); echo "$output"'` (quoted). Prefer `$(command)` over backticks for clarity and nesting.

Q166 How do you use sed for text processing?

Answer `'sed 's/old/new/g' file'` replaces all occurrences. `'s/old/new/'` replaces first per line. `'-i'` edits in-place (add backup: `'-i.bak'`). `'d'` deletes lines: `'sed '/pattern/d' file'`. `'p'` prints: `'sed -n '/pattern/p' file'`. Address ranges: `'sed '5,10d' file'` deletes lines 5-10. `'/start/,/end/d'` deletes range between patterns. `'sed 'n;d' file'` deletes every other line. `'sed -n '5p' file'` prints line 5.

Q167 How do you use awk for data processing?

Answer `awk` processes text field by field. '\$1,\$2' are fields (default delimiter: whitespace). '`awk '{print $1}' file`' prints first field. '`awk -F: '{print $1,$3}' /etc/passwd`' uses colon delimiter. '`awk 'NR>1{print}'`' skips header. '`awk '$3>100{print $1}'`' filters by condition. '`awk '{sum+=$2} END{print sum}'`' sums column. `BEGIN/END` blocks run before/after. '`awk '/pattern/{print}'`' filters by pattern.

Q168 How do you use `grep` effectively?

Answer '`grep pattern file`': basic search. '`-i`': case-insensitive. '`-r`': recursive directory search. '`-v`': invert (non-matching lines). '`-n`': show line numbers. '`-l`': list matching files only. '`-c`': count matches. '`-A 3`': show 3 lines after match. '`-B 3`': before. '`-C 3`': context (before and after). '`-E`': extended regex. '`-P`': Perl regex. '`-w`': whole word match. '`--include=*.log`': filter files. Multiple patterns: '`-e pattern1 -e pattern2`'.

Q169 What is the `xargs` command?

Answer `xargs` builds and executes commands from stdin input. '`find . -name "*.log" | xargs rm`' deletes all found log files. '`-I{}`' sets placeholder: '`ls *.txt | xargs -I{} cp {} /backup/`'. '`-n 2`' passes 2 arguments per command invocation. '`-P 4`' runs 4 parallel processes. '`--null`' or '`-0`' handles filenames with spaces (pair with '`find ... -print0`'). '`echo "hello" | xargs echo -n`' strips newlines.

Q170 How do you schedule scripts with `cron`?

Answer Edit with '`crontab -e`'. Format: 'min hour day month weekday command'. '`30 6 * * 1-5 /scripts/backup.sh >> /var/log/backup.log 2>&1`' runs at 6:30 AM weekdays. Always use full paths in cron scripts (no `PATH` inheritance). Redirect output to a log file. Test the script manually first. Set `MAILTO=""` to disable email output. '`crontab -l`' lists current jobs. System jobs: `/etc/cron.d/`.

Q171 What is the `trap` command in Bash?

Answer '`trap`' catches signals and executes code. '`trap "cleanup" EXIT`' runs `cleanup()` when script exits (any reason). '`trap "echo Interrupted" INT`' catches `Ctrl+C`. '`trap "" SIGTERM`' ignores `SIGTERM`. '`trap - INT`' resets to default. Common pattern: '`trap 'rm -f /tmp/lockfile' EXIT INT TERM`' ensures temp files are cleaned up. '`trap "echo Line $LINENO" ERR`' runs on error.

Q172 How do you create and use temporary files safely?

Answer '`mktemp`' creates a secure temporary file: '`tmpfile=$(mktemp)`'. '`mktemp -d`' creates a temp directory. Always use `mktemp` instead of hardcoded paths — prevents race conditions (TOCTOU vulnerabilities). Clean up: '`trap 'rm -f "$tmpfile"' EXIT`'. Set restrictive permissions if needed: '`chmod 600 "$tmpfile"`'. Temp files created by `mktemp` are owned by the running user.

Q173 How do you use process substitution in Bash?

Answer Process substitution creates a named pipe for command output. '<(command)' creates a file-like input: 'diff <(sort file1) <(sort file2)' compares sorted files without temp files. '>(command)' sends output to a command: 'tee >(gzip > out.gz) | process'. Useful for: 'while read line; do ...; done < <(command)' — avoids subshell that would lose variables (unlike pipe | while).

Q174 What is the difference between [] and [[]] in Bash?

Answer '[]' (test command): POSIX-compatible, available in all shells. Requires quoting variables. No regex. Pattern matching with case. '[[]]' (bash keyword): Bash-specific, more powerful. No word splitting inside — safe without quoting. Regex matching with =~: '[["\$var" =~ ^[0-9]+\$]]' . Logical operators && and || work directly (no need for -a/-o). Glob patterns: '[["\$file" == *.txt]]' . Prefer [[]] in Bash scripts.

Q175 How do you handle signals in a script?

Answer Signals tell processes to perform actions. Common signals: SIGTERM (15, graceful stop), SIGKILL (9, force kill), SIGHUP (1, reload config), SIGINT (2, Ctrl+C), SIGUSR1/2 (user-defined). 'trap "handler" SIGTERM SIGINT' catches multiple signals. 'kill -SIGUSR1 PID' sends user signal. 'signal_handler() { echo "Caught signal"; cleanup; exit 0; }; trap signal_handler SIGTERM'. SIGKILL cannot be caught.

Q176 What is the difference between source and executing a script?

Answer Executing ('bash script.sh' or './script.sh') runs script in a subshell — variables and directory changes don't affect the current shell. 'source script.sh' or '. script.sh' runs script in the current shell — changes DO affect current environment. Use source to load functions and variables into current shell (like .bashrc). Use source when script needs to change \$PWD or export variables that the parent needs.

Q177 How do you write portable shell scripts?

Answer Use '#!/bin/sh' (POSIX sh) if portability is needed, '#!/bin/bash' if Bash features required. Avoid Bash-specific features in POSIX scripts. Use 'command -v tool >/dev/null 2>&1 || { echo "tool required"; exit 1; }' to check dependencies. Use full paths for commands. Test with 'shellcheck' static analysis. Avoid platform-specific tools (use 'printf' instead of 'echo -e'). 'checkbashisms script.sh' finds bashisms in sh scripts.

Q178 What is shellcheck and why should you use it?

Answer : shellcheck is a static analysis tool for shell scripts that finds bugs, pitfalls, and style issues. It catches: unquoted variables, deprecated syntax, wrong redirections, masked errors, common mistakes. Install: 'apt install shellcheck'. Run: 'shellcheck script.sh'. Available online at shellcheck.net. Integrate into CI/CD pipelines. Most code editors have shellcheck plugins. Use for all shell scripts before deployment.

Q179 How do you handle configuration files in shell scripts?

Answer : Source a config file: '. /etc/myapp/config' or 'source "/etc/myapp/config"'. Config format (KEY=value, no spaces around =). Validate required variables: '\${VAR:? "VAR is required and not set"}'. Use defaults: '\${VAR:-default_value}'. Example pattern: 'CONFIG_FILE="/etc/myapp/config"; [-f "\$CONFIG_FILE"] && source "\$CONFIG_FILE"'. Protect sensitive configs: 'chmod 600 config; chown root:root config'.

Section 7: System Monitoring & Troubleshooting

Questions 181 – 210

Q181 How do you monitor system performance in Linux?

Answer Top-level tools: 'top' (interactive CPU/memory/process monitor), 'htop' (enhanced top), 'vmstat 1 5' (CPU, memory, I/O, swap every second), 'iostat -x 1' (disk I/O stats), 'sar' (historical performance from sysstat). 'dstat' combines multiple metrics. 'glances' provides comprehensive system overview. For long-term trending: collectd, Prometheus+node_exporter, or Nagios.

Q182 How do you identify a process consuming high CPU?

Answer 'top' then press 'P' to sort by CPU. 'ps aux --sort=-%cpu | head -10' lists top consumers. 'pidstat 1 5' shows per-process CPU (from sysstat). 'perf top' shows kernel-level CPU consumers for performance analysis. 'strace -p PID' traces system calls of a process. 'lsof -p PID' shows files opened. Once identified, check process purpose before killing.

Q183 How do you diagnose high memory usage?

Answer 'free -h' shows overall RAM/swap. 'cat /proc/meminfo' shows detailed breakdown (MemAvailable is key – not MemFree). 'ps aux --sort=-%mem | head -10' finds memory hogs. 'smem -s rss -r' shows PSS/RSS per process. 'pmap -x PID' shows process memory map. OOM killer events: 'dmesg | grep -i oom' or 'grep oom /var/log/syslog'. Enable transparent huge pages: check /sys/kernel/mm/transparent_hugepage/.

Q184 How do you diagnose high disk I/O?

Answer 'iostat -x 1' shows per-device I/O: look at %util (saturation), await (latency), r/s w/s (read/write ops). 'iotop -o' shows processes actively doing I/O. 'lsof +D /path' finds processes with open files in a directory. 'hdparm -tT /dev/sda' benchmarks disk speed. 'blktrace' provides block-level I/O tracing. 'atop' combines process and disk stats. High iowait in CPU stats indicates disk bottleneck.

Q185 What does the vmstat command show?

Answer 'vmstat 1 5' shows 5 samples every 1 second. Key columns: procs: r (running), b (blocked in sleep). memory: swpd (swap used), free, buff, cache. swap: si/so (swap in/out – non-zero means RAM pressure). io: bi/bo (block in/out). system: in (interrupts), cs (context switches). cpu: us (user), sy (system), id (idle), wa (iowait), st (stolen by hypervisor).

Q186 How do you troubleshoot boot problems?

Answer If system won't boot: 1) GRUB menu — edit kernel params, add 'rd.break' for dracut shell or 'init=/bin/bash' for minimal shell. 2) Check last successful boot: 'journalctl -b -1'. 3) Boot to rescue/emergency target. 4) Check 'dmesg' for hardware errors. 5) Check '/var/log/boot.log'. 6) Run 'fsck' on filesystems from live media. 7) Check '/etc/fstab' for invalid entries. 'systemctl list-units --failed' shows failed units.

Q187 How do you find and kill zombie processes?

Answer Zombie processes (defunct) have completed but parent hasn't called wait(). Show: 'ps aux | grep Z' or 'ps -eo pid,ppid,stat,cmd | grep Z'. Zombies can't be killed directly — kill the parent: 'kill -SIGCHLD PPID' signals parent to reap children. If parent is unresponsive, 'kill PPID'. Zombies accumulate if parent is buggy. They consume no resources (just PID slot) until reaped.

Q188 What is the /proc/sys/ directory used for?

Answer /proc/sys/ exposes kernel tunables as files. Read: 'cat /proc/sys/net/ipv4/ip_forward'. Write: 'echo 1 > /proc/sys/net/ipv4/ip_forward'. Equivalent sysctl: 'sysctl -w net.ipv4.ip_forward=1'. Important tunables: vm.swappiness (swap aggressiveness), net.ipv4.tcp_syncookies (SYN flood protection), kernel.panic (auto-reboot on panic), fs.file-max (max open files system-wide), vm.overcommit_memory.

Q189 How do you trace system calls of a process?

Answer 'strace command' traces all system calls as command runs. 'strace -p PID' attaches to running process. '-e trace=open,read,write' filters specific calls. '-o output.txt' writes to file. '-f' follows forked children. '-T' shows time spent in each call. '-c' provides summary statistics. 'ltrace' traces library calls. Both are invaluable for debugging application behavior and permission issues.

Q190 How do you diagnose network connectivity issues?

Answer Systematic: 1) 'ip link show' — physical/driver layer up? 2) 'ip addr show' — IP assigned? 3) 'ping 127.0.0.1' — loopback OK. 4) 'ping default_gateway' — local routing works? 5) 'ping 8.8.8.8' — internet reachable (no DNS)? 6) 'dig google.com' or 'nslookup google.com' — DNS resolves? 7) 'traceroute host' — where does routing fail? 8) 'ss -tln' — is service listening? 9) 'curl -v URL' — application layer.

Q191 What is the netstat command and what has replaced it?

Answer 'netstat' shows network connections, routing tables, interface stats, and sockets. Common flags: '-tuln' (listening TCP/UDP), '-an' (all connections, numeric), '-rn' (routing table). Deprecated in favor of 'ss' (faster, more info) for connections, 'ip route' for routing, 'ip -s link' for interface stats. 'ss -tuln' is the direct replacement for 'netstat -tuln'. Most Linux distros still include netstat via net-tools package.

Q192 How do you check open files and file handles?

Answer 'lsof' (list open files) shows all open files. 'lsof -u username': files opened by user. 'lsof -p PID': files opened by process. 'lsof /path': processes using a path. 'lsof -i :80': processes using port 80. 'lsof -i TCP': all TCP connections. 'lsof +D /mnt/data': all files under a mount (why you can't unmount). 'cat /proc/sys/fs/file-nr': system-wide file handle usage and limits.

Q193 What is the purpose of /sys/class/net/?

Answer /sys/class/net/ contains directories for each network interface with detailed attributes. 'cat /sys/class/net/eth0/speed' shows link speed. 'cat /sys/class/net/eth0/carrier' shows if link is up (1) or down (0). 'cat /sys/class/net/eth0/statistics/rx_bytes' shows received bytes. 'cat /sys/class/net/eth0/operstate' shows operational state. Useful for monitoring and scripting without parsing command output.

Q194 How do you check hardware information in Linux?

Answer 'lshw' shows detailed hardware inventory (-short for summary, -class disk for specific). 'lscpu' shows CPU information. 'lspci' shows PCI devices (graphics, network, storage controllers). 'lsusb' shows USB devices. 'dmidecode' reads BIOS/UEFI/DMI info (memory slots, system info). 'hwinfo' provides comprehensive hardware detection. 'inxi -Fxz' gives a comprehensive system info summary.

Q195 What is the sar command?

Answer sar (System Activity Reporter) from sysstat collects, reports, and saves system activity data. 'sar -u 1 5': CPU usage every second, 5 times. 'sar -r 1 5': memory. 'sar -d 1 5': disk activity. 'sar -n DEV 1': network. 'sar -q 1': load average. Historical data (by default collected hourly) stored in /var/log/sa/. 'sar -u -f /var/log/sa/sa15' reads data for the 15th. Enable with 'systemctl enable --now sysstat'.

Q196 How do you check for hardware errors?

Answer `'dmesg -T | grep -i error'` shows recent kernel hardware errors. `'mcelog'` logs machine check exceptions (CPU/memory hardware errors). `'smartctl -a /dev/sda'` (from `smartmontools`) shows S.M.A.R.T. disk health data — look for reallocated sectors. `'memtester 1G 2'` tests RAM. `'badblocks -v /dev/sdb'` scans for bad blocks (only on unmounted filesystems). Check `/var/log/mcelog` for hardware errors.

Q197 What is the ping command and how do you interpret it?

Answer `'ping -c 4 host'` sends 4 ICMP echo requests. Output shows: bytes, icmp_seq, TTL (time-to-live — hops remaining), and round-trip time in ms. Summary: packets transmitted, received, packet loss %, min/avg/max/stddev RTT. High RTT indicates latency. Packet loss indicates connectivity problems. `'ping -f host'` (flood ping) for stress testing. Blocked ping doesn't necessarily mean host is down.

Q198 How do you identify what's listening on a specific port?

Answer `'ss -tlnp | grep :80'` shows process listening on port 80. `'fuser 80/tcp'` shows PID using port. `'lsof -i :80'` shows process details. `'netstat -tlnp | grep :80'` (if netstat available). In `/proc`: `'cat /proc/net/tcp'` shows raw TCP connections (ports in hex). `'systemctl list-units --type=socket'` shows systemd socket units. nmap localhost can also scan locally.

Q199 How do you troubleshoot DNS resolution?

Answer `'dig domain.com'` performs DNS lookup with detailed output. `'dig @8.8.8.8 domain.com'` tests specific DNS server. `'dig -x 1.2.3.4'` reverse lookup. `'nslookup domain.com'` basic lookup. `'host domain.com'` simple resolution. Check `/etc/resolv.conf` for nameservers. Check `/etc/nsswitch.conf` for resolution order. `'systemd-resolve --status'` or `'resolvectl status'` shows systemd-resolved config. `'dig +trace domain.com'` traces the full resolution path.

Q200 How do you monitor disk health with SMART?

Answer `smartmontools` provides SMART monitoring. `'smartctl -i /dev/sda'` shows device info. `'smartctl -H /dev/sda'` shows health (PASSED/FAILED). `'smartctl -a /dev/sda'` shows all SMART data. Key attributes: `Reallocated_Sector_Ct` (non-zero is bad), `Pending_Sector_Count`, `Uncorrectable_Sector_Count`. Run short test: `'smartctl -t short /dev/sda'`. Enable background monitoring: `'systemctl enable smartmontools'`. Alert on failures.

Q201 What is the purpose of the /sys filesystem?

Answer /sys (sysfs) is a virtual filesystem exposing the kernel's device model. It provides structured access to: devices (/sys/devices/), drivers (/sys/bus/*/drivers/), kernel modules (/sys/module/), block devices (/sys/class/block/), network interfaces (/sys/class/net/), power management (/sys/power/). Unlike /proc, sysfs follows a hierarchical model matching the actual device tree. Writable files allow runtime configuration.

Q202 How do you check and configure system limits?

Answer 'ulimit -a' shows current shell limits. 'ulimit -n 65535' sets open files limit for current session. Persistent limits in /etc/security/limits.conf: 'username soft nofile 10000' and 'hard nofile 65535'. For services, add LimitNOFILE=65535 in [Service] section of systemd unit. 'cat /proc/PID/limits' shows limits for a specific process. System-wide: /etc/sysctl.conf 'fs.file-max = 2097152'.

Q203 What is the purpose of the dstat command?

Answer 'dstat' provides a versatile replacement for vmstat, iostat, netstat, and ifstat in one tool. 'dstat' shows CPU, disk, net, paging, and system stats together. 'dstat -cdngm' shows CPU, disk, net, net errors, memory. 'dstat --top-cpu' shows highest CPU process. 'dstat -ta --output report.csv' saves to CSV for analysis. 'dstat 1 30' shows 30 samples at 1-second intervals. Requires dstat package.

Q204 How do you use atop for system monitoring?

Answer 'atop' provides advanced system monitoring with process and resource accounting. It shows CPU, memory, disk, and network per process. Logs system activity for playback: 'atop -r /var/log/atop/atop_YYYYMMDD' to analyze past performance. 'atop -w file' writes to file; 'atop -r file' reads back. The atop daemon logs every 10 minutes by default. Press 'm' for memory sort, 'd' for disk, 'c' for command line view.

Q205 How do you diagnose a hung or unresponsive system?

Answer If you have remote/console access: 1) 'w' and 'top' to see load and processes. 2) Check 'dmesg' for kernel panics or OOM kills. 3) 'ps -eo pid,stat,cmd | grep " D "' finds processes in uninterruptible sleep (disk/NFS issue). 4) 'lsof | grep deleted' finds deleted files holding disk space. 5) Check /var/log/messages or journalctl. 6) Magic SysRq key for emergency: Alt+SysRq+T (backtrace), Alt+SysRq+K (kill), Alt+SysRq+B (reboot).

Q206 What is the difference between soft and hard limits?

Answer : Soft limits: current effective limit, can be increased by user up to the hard limit. 'ulimit -Sn 5000' sets soft open files limit. Hard limits: ceiling set by root, unprivileged users can't exceed. 'ulimit -Hn 65535' sets hard limit (root only). View all: 'ulimit -aS' (soft) and 'ulimit -aH' (hard). In /etc/security/limits.conf: 'user soft nproc 1000' and 'user hard nproc 2000'.

Q207 How do you check swap usage and manage it?

Answer : 'free -h' shows swap total, used, free. 'swapon -s' lists active swap devices with priority and usage. 'cat /proc/swaps' raw swap info. 'vmstat 1' shows si/so (swap in/out rate — high values mean memory pressure). 'sysctl vm.swappiness' shows aggressiveness (default 60, lower = prefer RAM). 'swapoff -a && swapon -a' recycles swap (use with caution). 'smem' shows per-process actual RAM + swap usage.

Q208 What is the load average and how do you interpret it?

Answer : Load average shows average processes in run queue over 1, 5, 15 minutes (from 'uptime' and 'top'). On a 4-core system: load of 4.0 = 100% CPU utilization. Load > core count means queuing/overload. Check core count: 'nproc' or 'grep -c processor /proc/cpuinfo'. Load includes both CPU-bound and I/O-blocked processes. Spike in 1-min but low 15-min = brief spike. Sustained high 15-min = ongoing problem. Low CPU but high load = I/O bottleneck.

Q209 How do you use perf for performance profiling?

Answer : 'perf stat command' shows CPU performance counters for a command (cache misses, cycles, instructions). 'perf top' shows real-time top CPU functions (like top but for function calls). 'perf record -g -p PID' records call graphs. 'perf report' analyzes recorded data interactively. 'perf trace command' traces system calls (like strace but lower overhead). Requires kernel debug symbols for meaningful function names. Good for deep CPU performance analysis.

Section 8: Advanced Topics & Virtualization

Questions 211 – 240

Q211 What is KVM and how do you manage virtual machines?

Answer KVM (Kernel-based Virtual Machine) is a Linux hypervisor built into the kernel. Install: `apt install qemu-kvm libvirt-daemon virt-manager`. Check support: `grep -E "(vmx|svm)" /proc/cpuinfo`. Create VM: `virt-install --name vm1 --ram 2048 --disk size=20 --cdrom ubuntu.iso`. Manage with virsh: `virsh list --all` (list VMs), `virsh start/shutdown vm1`, `virsh console vm1`. virt-manager provides GUI management.

Q212 What are Linux namespaces?

Answer Namespaces provide process isolation – each namespace type isolates different resources. Types: pid (process IDs), net (network interfaces), mnt (mount points), uts (hostname), ipc (interprocess communication), user (user/group IDs), cgroup (cgroup root). Containers use namespaces for isolation. `lsns` lists namespaces. `unshare --pid --fork bash` creates new PID namespace. `nsenter -t PID --pid --net bash` enters a process's namespace.

Q213 What are cgroups (control groups)?

Answer Cgroups limit, account for, and isolate resource usage of process groups. Resources: cpu, memory, blkio (I/O), network (via net_cls). v1: `/sys/fs/cgroup/`. v2: unified hierarchy at `/sys/fs/cgroup/`. `cgcreate -g memory:mygroup` creates a cgroup. Set limit: `echo 512M > /sys/fs/cgroup/memory/mygroup/memory.limit_in_bytes`. systemd uses cgroups for service resource limits. `systemd-cgtop` shows cgroup resource usage.

Q214 How does Docker use Linux features?

Answer Docker containers use: namespaces (process isolation), cgroups (resource limits), union filesystems (overlay2 for layered images), seccomp profiles (syscall filtering), capabilities (drop unneeded root caps), and network bridges/namespaces for connectivity. `docker run --memory=512m --cpus=1 image` sets resource limits. Understanding these primitives helps troubleshoot container issues. `docker inspect container` shows all Linux-level settings.

Q215 What is systemd-nspawn?

Answer `systemd-nspawn` is a lightweight container manager similar to `chroot` but with namespace isolation. '`systemd-nspawn -D /path/to/rootfs`' starts a container from a directory. '`systemd-nspawn --machine=mycontainer -D /path/ -b`' boots a full `systemd` in the container. '`machinectl list`' shows running containers. Good for: testing, OS chroots, lightweight containers without full Docker stack. Part of `systemd` package.

Q216 What is the `chroot` command and when do you use it?

Answer '`chroot /newroot /bin/bash`' changes apparent root directory for a process, isolating it from the real filesystem. Use cases: 1) System rescue — boot from live media, `chroot` to broken system to fix bootloader or reinstall packages. 2) Building software in an isolated environment. 3) Running older software requiring different library versions. '`arch-chroot`' (on Arch) handles bind mounts automatically. '`systemd-nspawn`' is a more capable alternative.

Q217 How do you manage containers with `podman`?

Answer `Podman` is a daemonless OCI container tool, compatible with Docker commands. '`podman run -d --name web -p 80:80 nginx`' runs Nginx. '`podman ps`' lists running containers. '`podman images`' lists images. '`podman pull image`' downloads image. '`podman stop/start/rm name`'. '`podman exec -it container bash`' gets shell. Rootless: run containers as non-root user for better security. '`podman generate systemd container`' creates `systemd` unit.

Q218 What is Ansible and how does it work?

Answer Ansible is an agentless automation tool using SSH and Python. Inventory file defines hosts. Playbooks (YAML) define tasks. '`ansible all -i inventory -m ping`' tests connectivity. '`ansible-playbook -i inventory playbook.yml`' runs a playbook. Modules: `apt`, `yum`, `service`, `copy`, `template`, `user`, `file`. Roles organize complex playbooks. Idempotent: safe to run multiple times. No agent needed on managed nodes — just SSH and Python.

Q219 What is a `systemd` service unit file structure?

Answer [Unit]: `Description=`, `After=` (ordering), `Requires=`/`Wants=` (dependencies), `ConditionPathExists=`. [Service]: `Type=` (simple/forking/oneshot/notify), `ExecStart=`, `ExecStop=`, `ExecReload=`, `Restart=on-failure`, `RestartSec=5`, `User=`, `Group=`, `WorkingDirectory=`, `EnvironmentFile=`, `LimitNOFILE=`. [Install]: `WantedBy=multi-user.target`. Place in `/etc/systemd/system/`. '`systemctl daemon-reload`' after changes.

Q220 What are `systemd` socket activation?

Answer Socket activation allows systemd to listen on a socket and start the service only when a connection arrives — saves memory when services aren't busy. Create pairs: myservice.socket + myservice.service. In .socket: [Socket] ListenStream=80. In .service: [Service] ExecStart=... The service receives the socket as fd 3. 'systemctl enable myservice.socket' instead of service. Supported by inetd-style services. Enables parallel startup.

Q221 How do you create a custom systemd service?

Answer 1) Create /etc/systemd/system/myapp.service: [Unit] Description=My App After=network.target, [Service] Type=simple ExecStart=/usr/local/bin/myapp --config /etc/myapp.conf Restart=always User=myapp EnvironmentFile=/etc/myapp/env, [Install] WantedBy=multi-user.target. 2) 'systemctl daemon-reload'. 3) 'systemctl enable --now myapp'. 4) 'systemctl status myapp'. 5) 'journalctl -u myapp -f' for logs.

Q222 What is the difference between Type=simple and Type=forking in systemd?

Answer Type=simple: ExecStart process IS the main process. systemd considers service ready immediately. Best for programs that stay in foreground. Type=forking: ExecStart forks and parent exits — traditional daemon behavior. Requires PIDFile= so systemd tracks the child. Type=notify: service sends sd_notify() when ready — most reliable. Type=oneshot: runs and exits, for scripts. Type=idle: like simple but waits until other jobs done.

Q223 What are Linux kernel modules?

Answer Kernel modules extend kernel functionality without recompilation. 'lsmod' lists loaded modules. 'modprobe modulename' loads a module (with dependencies). 'modprobe -r modulename' removes. 'insmod module.ko' loads without dependencies. 'rmmod modulename' removes. 'modinfo modulename' shows module info. '/etc/modules' (Debian) or '/etc/modules-load.d/' (systemd) loads modules at boot. '/etc/modprobe.d/' contains module options.

Q224 How do you configure network interfaces at boot persistently?

Answer Method depends on distro/tool: NetworkManager: 'nmcli connection modify ...' (persists to /etc/NetworkManager/system-connections/). systemd-networkd: create /etc/systemd/network/10-eth0.network with [Match] and [Network] sections. Debian /etc/network/interfaces: 'auto eth0 iface eth0 inet static address 192.168.1.10 netmask 255.255.255.0'. RHEL /etc/sysconfig/network-scripts/ifcfg-eth0: 'BOOTPROTO=none IPADDR=192.168.1.10'.

Q225 What is the difference between /etc/sysconfig and /etc/default?

Answer `/etc/sysconfig/` (Red Hat/RHEL/CentOS/Fedora): stores service configuration variables. Example: `/etc/sysconfig/network` (network settings), `/etc/sysconfig/selinux` (SELinux config), `/etc/sysconfig/docker` (Docker options). `/etc/default/` (Debian/Ubuntu): serves same purpose — service default variables. Example: `/etc/default/grub`, `/etc/default/docker`, `/etc/default/nfs-common`. Scripts source these files for configurable options.

Q226 How do you configure systemd resource limits for services?

Answer In `[Service]` section of unit file: `LimitNOFILE=65535` (max open files). `LimitNPROC=1000` (max processes). `LimitMEMLOCK=infinity` (locked memory). `CPUQuota=50%` (limit to 50% CPU). `MemoryLimit=512M` (memory limit). `BlockIOWeight=100` (I/O priority). `TasksMax=512`. After adding: `'systemctl daemon-reload && systemctl restart service'`. Verify: `'systemctl show service | grep Limit'`. `'cat /proc/PID/limits'` confirms limits are applied.

Q227 What is the XFS filesystem and how do you manage it?

Answer XFS is a high-performance journaling filesystem (default on RHEL 7+). Create: `'mkfs.xfs /dev/sdb1'`. Grow online: `'xfs_growfs /mount/point'` (can't shrink). Repair: `'xfs_repair /dev/sdb1'` (must be unmounted). Freeze/thaw (for snapshots): `'xfs_freeze -f /mount'` and `'xfs_freeze -u /mount'`. Check: `'xfs_info /dev/sdb1'`. Defragment: `'xfs_fsr /mount'`. XFS handles large files and parallel I/O very well.

Q228 What is the btrfs filesystem and its key management commands?

Answer btrfs management: `'btrfs filesystem show'` lists btrfs filesystems. `'btrfs filesystem df /mnt'` shows space usage. `'btrfs subvolume create /mnt/subvol'` creates subvolume. `'btrfs subvolume list /mnt'` lists. `'btrfs subvolume snapshot /mnt/data /mnt/snap01'` creates snapshot. `'btrfs send /mnt/snap | ssh host btrfs receive /mnt/'` replicates. `'btrfs balance start /mnt'` rebalances data. `'btrfs scrub start /mnt'` checks data integrity.

Q229 How do you configure bonding in NetworkManager?

Answer `'nmcli connection add type bond con-name bond0 ifname bond0'` creates bond. `'nmcli connection modify bond0 bond.options "mode=active-backup,miimon=100"'` sets mode. Add slaves: `'nmcli connection add type ethernet con-name bond0-slave0 ifname eth0 master bond0'`. `'nmcli connection add type ethernet con-name bond0-slave1 ifname eth1 master bond0'`. `'nmcli connection up bond0'`. `'cat /proc/net/bonding/bond0'` shows bond status.

Q230 What is the difference between a virtual machine and a container?

Answer VMs: full hardware virtualization, each VM runs its own OS kernel, strong isolation, higher overhead (GB RAM, seconds to start), managed by hypervisors (KVM, VMware). Containers: OS-level virtualization, share host kernel, lighter isolation (namespaces/cgroups), low overhead (MB RAM, milliseconds to start), managed by Docker/Podman/LXC. VMs for: different OSes, strong isolation, legacy apps. Containers for: microservices, rapid deployment, dev/prod parity.

Q231 What is LXC/LXD?

Answer LXC (Linux Containers) provides OS-level virtualization using namespaces and cgroups, offering a full Linux system environment without a separate kernel. LXD is a management layer on top of LXC with better tooling. 'lxc launch ubuntu:20.04 mycontainer' creates a container. 'lxc exec mycontainer bash' gets shell. 'lxc list' shows containers. 'lxc snapshot mycontainer snap01' takes snapshot. More VM-like than Docker — runs full init system.

Q232 How do you configure iSCSI initiator?

Answer 1) Install: 'apt install open-iscsi' or 'yum install iscsi-initiator-utils'. 2) 'iscsiadm -m discovery -t sendtargets -p target_ip' discovers targets. 3) 'iscsiadm -m node -T iqn.target.name -p target_ip --login' connects. 4) Block device appears as /dev/sdX. 5) Format and mount normally. 6) Persistent: 'iscsiadm -m node -T iqn.target -p ip --op update -n node.startup -v automatic'. Check sessions: 'iscsiadm -m session'.

Q233 What is the difference between ext4 and XFS for databases?

Answer For databases, XFS is generally preferred: better parallel I/O scalability, handles many small concurrent I/O operations well, allocates large extents efficiently, no fragmentation issues with large databases. ext4 is simpler and reliable but can struggle with many concurrent writes at scale. Both support direct I/O (bypassing page cache) needed by some databases. Always disable atime ('noatime' mount option) for database filesystems regardless of choice.

Q234 How do you use the nmap command for security auditing?

Answer 'nmap -sV host' scans and identifies service versions. 'nmap -O host' OS detection. 'nmap -p 1-65535 host' scans all ports. 'nmap -sn 192.168.1.0/24' ping sweep (host discovery). 'nmap -A host' aggressive (OS, version, scripts, traceroute). 'nmap --script vuln host' runs vulnerability scripts. 'nmap -sS' SYN scan (stealth). 'nmap -6 host' IPv6 scan. Always have authorization before scanning — unauthorized scanning may be illegal.

Q235 What is the purpose of the /proc/PID/ directory?

Answer Each running process has a directory `/proc/PID/` with its runtime information. Key files: `cmdline` (full command), `environ` (environment variables), `fd/` (file descriptors – symlinks to open files), `maps` (memory mappings), `status` (state, memory, PIDs), `limits` (resource limits), `net/` (network sockets), `exe` (symlink to executable), `cwd` (symlink to current directory), `io` (I/O statistics). Used by `ps`, `top`, and monitoring tools to gather process info.

Q236 How do you configure NFS server and client?

Answer Server: 1) `'apt install nfs-kernel-server'`. 2) Edit `/etc/exports`: `'/data 192.168.1.0/24(rw,sync,no_subtree_check,no_root_squash)'`. 3) `'exportfs -ra'` applies exports. 4) `'systemctl enable --now nfs-server'`. Client: 1) `'apt install nfs-common'`. 2) `'mount -t nfs server:/data /mnt/data'`. 3) Persistent `fstab`: `'server:/data /mnt/data nfs defaults,_netdev 0 0'`. `'showmount -e server'` lists exports. `'exportfs -v'` shows current exports.

Q237 What is Samba and how do you configure a share?

Answer Samba provides SMB/CIFS file sharing for Windows interoperability. Install: `'apt install samba'`. Config: `/etc/samba/smb.conf`. Add share: `'[myshare] path = /srv/share valid users = alice writable = yes'`. `'testparm'` validates config. Set Samba password: `'smbpasswd -a alice'`. Restart: `'systemctl restart smbd nmbd'`. Client mount: `'mount -t cifs //server/myshare /mnt -o user=alice'`. `'smbclient -L //server'` lists shares.

Q238 How do you configure a DHCP server?

Answer Install `isc-dhcp-server`: `'apt install isc-dhcp-server'`. Config `/etc/dhcp/dhcpd.conf`: `'subnet 192.168.1.0 netmask 255.255.255.0 { range 192.168.1.100 192.168.1.200; option routers 192.168.1.1; option domain-name-servers 8.8.8.8; default-lease-time 86400; max-lease-time 172800; }'`. Set interface: `INTERFACES=eth0` in `/etc/default/isc-dhcp-server`. `'systemctl enable --now isc-dhcp-server'`. Leases: `/var/lib/dhcp/dhcpd.leases`.

Q239 What is the Logical Volume Manager thin provisioning?

Answer Thin provisioning creates volumes that appear larger than physically allocated space – space is allocated from a pool only when data is written. Create thin pool: `'lvcreate --thinpool mypool -L 100G datavg'`. Create thin volume: `'lvcreate --thin -n thinvol --virtualsize 500G datavg/mypool'`. Format and mount normally. Monitor: `'lvdisplay datavg/mypool'` shows data/metadata % used. Over-provisioning is possible but must be monitored to prevent pool exhaustion.

Section 9: Kernel & Boot Management

Questions 241 – 270

Q241 What is the Linux boot process in detail?

Answer 1) Power on: CPU executes BIOS/UEFI firmware. 2) BIOS POST (Power-On Self Test): hardware check. 3) Bootloader selection: BIOS reads MBR; UEFI reads EFI System Partition. 4) GRUB2 loads kernel image (vmlinuz) and initramfs into RAM. 5) Kernel decompresses, initializes hardware, mounts initramfs as temporary /. 6) Kernel executes /sbin/init (PID 1) – systemd on modern systems. 7) systemd reads unit files, starts targets, services, and login manager.

Q242 What is GRUB2 and how do you configure it?

Answer GRUB2 is the standard bootloader. Config generated by 'grub-mkconfig' or 'update-grub' from /etc/default/grub and scripts in /etc/grub.d/. Never edit /boot/grub/grub.cfg directly. Key /etc/default/grub settings: GRUB_TIMEOUT=5 (menu timeout), GRUB_DEFAULT=0 (default entry), GRUB_CMDLINE_LINUX_DEFAULT="quiet splash" (kernel params). After editing: 'update-grub' (Debian) or 'grub2-mkconfig -o /boot/grub2/grub.cfg' (RHEL).

Q243 How do you recover from a broken GRUB bootloader?

Answer From GRUB rescue prompt: 'ls' (list devices), 'set root=(hd0,1)', 'linux /vmlinuz root=/dev/sda1', 'initrd /initrd.img', 'boot'. From live media: 1) Boot live USB. 2) Mount root: 'mount /dev/sda1 /mnt'. 3) Bind mounts: 'mount --bind /dev /mnt/dev; mount --bind /proc /mnt/proc; mount --bind /sys /mnt/sys'. 4) 'chroot /mnt'. 5) 'grub-install /dev/sda'. 6) 'update-grub'. 7) Reboot.

Q244 What are kernel boot parameters?

Answer Common kernel parameters: 'quiet' (suppress messages), 'splash' (show splash screen), 'ro/rw' (mount root read-only/read-write), 'root=UUID=xxx' (root device), 'init=/bin/bash' (bypass init for recovery), 'single' or '1' (single-user mode), 'rd.break' (break at initramfs early userspace), 'nomodeset' (disable kernel mode setting – fix graphics issues), 'ipv6.disable=1' (disable IPv6), 'mem=2G' (limit RAM). Edit in GRUB menu at boot.

Q245 What is initramfs and how do you rebuild it?

Answer initramfs (initial RAM filesystem) is a compressed cpio archive loaded into RAM at boot. Contains drivers, tools, and scripts needed to mount the real root filesystem (especially for LVM, RAID, encryption, or exotic filesystems). Rebuild: 'update-initramfs -u' (Debian/Ubuntu) or 'dracut -f' (RHEL). 'update-initramfs -u -k all' rebuilds for all installed kernels. 'lsinitramfs /boot/initrd.img' lists contents (Debian).

Q246 What is dracut?

Answer dracut is the initramfs infrastructure used by RHEL/CentOS/Fedora (and optionally other distros). It creates initramfs on demand. 'dracut -f' rebuilds for current kernel. 'dracut -f /boot/initramfs-\$(uname -r).img \$(uname -r)' rebuilds explicitly. 'dracut --list-modules' shows available modules. Config: /etc/dracut.conf and /etc/dracut.conf.d/. 'lsinitrd /boot/initramfs-\$(uname -r).img | head -20' lists initrd contents.

Q247 How do you install a new kernel?

Answer Package manager installs new kernels automatically. On Debian: 'apt install linux-image-5.15.0-76-generic'. On RHEL: 'dnf install kernel-5.14.0'. Old kernels remain – GRUB lists all. 'uname -r' shows running kernel. 'dpkg --get-selections | grep linux-image' (Debian) or 'rpm -qa kernel' (RHEL) lists installed kernels. Remove old: 'apt autoremove' or 'package-cleanup --oldkernels --count=2' (RHEL). GRUB updates automatically.

Q248 How do you configure kdump for kernel crash dumps?

Answer kdump captures kernel crash dumps using kexec (boots a second 'capture' kernel on crash). Install: 'apt install kdump-tools' or 'yum install kexec-tools'. Reserve memory: add 'crashkernel=auto' to kernel params in /etc/default/grub, then update-grub. Enable: 'systemctl enable --now kdump'. Configure dump location in /etc/kdump.conf: 'path /var/crash'. After crash, analyze with: 'crash /boot/vmlinuz-\$(uname -r) /var/crash/vmcore'.

Q249 What is PXE boot and how do you set it up?

Answer PXE (Preboot Execution Environment) boots computers from network. Setup: 1) DHCP server with next-server (TFTP) and filename (pxelinux.0) options. 2) TFTP server serving bootloader and kernel. 3) NFS/HTTP server serving OS files. 4) PXE config in /var/lib/tftpboot/pxelinux.cfg/. Used for: network OS installations, diskless workstations, and automated provisioning (Kickstart/Preseed). 'cobbler' is a popular tool for PXE deployment automation.

Q250 What are systemd targets and how do you use them?

Answer Targets group units that should be activated together. Common targets: poweroff.target, rescue.target, multi-user.target, graphical.target, reboot.target, emergency.target. 'systemctl get-default' shows default. 'systemctl set-default multi-user.target' changes default. 'systemctl isolate rescue.target' switches immediately. List all: 'systemctl list-units --type=target'. Dependencies declared with Wants=, Requires=, Before=, After= in unit files.

Q251 How does UEFI Secure Boot work?

Answer Secure Boot is a UEFI feature that only allows software signed with trusted keys to run at boot. Key database: PK (Platform Key), KEK (Key Exchange Key), db (allowed), dbx (forbidden/revoked). Linux uses a 'shim' bootloader signed by Microsoft, which then verifies the distro's GRUB, which verifies the kernel. 'mokutil --sb-state' shows Secure Boot state. 'mokutil --import key.der' enrolls a custom key (MOK = Machine Owner Key).

Q252 What is UEFI and how does it differ from BIOS?

Answer UEFI (Unified Extensible Firmware Interface) replaces legacy BIOS. Differences: UEFI supports GPT disks (>2TB), BIOS only MBR. UEFI has Secure Boot, BIOS doesn't. UEFI stores bootloaders in EFI System Partition (ESP, typically /boot/efi, FAT32), BIOS reads MBR. UEFI provides graphical interface and mouse support. UEFI boots faster (parallel initialization). UEFI can boot 64-bit directly; BIOS needs 16-bit real mode first.

Q253 How do you manage UEFI boot entries?

Answer 'efibootmgr' manages UEFI boot entries. 'efibootmgr -v' lists entries with details. 'efibootmgr -o 0001,0000' changes boot order. 'efibootmgr -n 0001' boots specific entry next time only. 'efibootmgr -b 0001 -B' deletes an entry. 'efibootmgr -c -d /dev/sda -p 1 -L "Ubuntu" -l /EFI/ubuntu/grubx64.efi' creates new entry. ESP (EFI System Partition) is typically mounted at /boot/efi.

Q254 What is the difference between GRUB legacy and GRUB2?

Answer GRUB legacy (0.9x): config in /boot/grub/menu.lst (hand-edited). GRUB2 (1.9x+, now 2.x): config auto-generated from /etc/default/grub + /etc/grub.d/ scripts by grub-mkconfig. GRUB2 supports: GPT, UEFI, encrypted partitions, LVM, Btrfs subvolumes, scripting in config. Commands differ: 'root (hd0,0)' becomes 'set root=(hd0,1)' (GPT uses 1-based partition numbers). GRUB2 is standard on all modern distros.

Q255 How do you configure the kernel for a specific workload?

Answer Key sysctl tunables by workload: Database: 'vm.swappiness=10', 'vm.dirty_ratio=5'. Web server: 'net.core.somaxconn=65535', 'net.ipv4.tcp_max_syn_backlog=65535'. HPC: 'kernel.sched_migration_cost_ns=5000000', transparent hugepages=always. Container host: 'net.bridge.bridge-nf-call-iptables=1', 'user.max_inotify_watches=1048576'. Set permanently in /etc/sysctl.d/99-custom.conf. 'sysctl -p' applies.

Q256 What is a kernel panic and how do you investigate it?

Answer A kernel panic is a fatal system error the kernel can't recover from — system halts or reboots. Causes: hardware failure, driver bugs, file system corruption, memory errors. Investigate: 'journalctl -b -1' (previous boot logs), 'dmesg | grep -i panic', /var/log/messages. Configure automatic reboot: 'sysctl -w kernel.panic=30' (reboot after 30s). Enable kdump for crash dumps. Check S.M.A.R.T. for disk errors. 'mcelog' for CPU/memory hardware errors.

Q257 What is the /boot directory structure?

Answer /boot contains files needed for system boot: vmlinuz-VERSION (compressed kernel image), initrd.img-VERSION or initramfs-VERSION.img (initial RAM disk), System.map-VERSION (kernel symbol table for debugging), config-VERSION (kernel build configuration), grub/ (GRUB2 files including grub.cfg). On UEFI: /boot/efi/EFI/ contains bootloader files. 'ls -lh /boot' shows sizes — ensure /boot has sufficient free space (typically 250MB+).

Q258 How do you update the system bootloader?

Answer GRUB2 (Debian/Ubuntu): 'update-grub' regenerates /boot/grub/grub.cfg. 'grub-install /dev/sda' reinstalls GRUB to MBR. GRUB2 (RHEL): 'grub2-mkconfig -o /boot/grub2/grub.cfg' (BIOS) or 'grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg' (UEFI). 'grub2-install /dev/sda' reinstalls. After kernel param changes in /etc/default/grub, always run update-grub/grub2-mkconfig to apply changes.

Q259 What is the SysRq (Magic Keys) feature?

Answer SysRq allows issuing low-level commands directly to the kernel, bypassing normal operations. Enable: 'sysctl -w kernel.sysrq=1' or 'echo 1 > /proc/sys/kernel/sysrq'. Keyboard: Alt+PrtScn+key. Via proc: 'echo key > /proc/sysrq-trigger'. Key commands: b (reboot immediately), c (crash for kdump), k (kill all processes on console), s (sync filesystems), u (remount read-only), t (thread dump to console), m (memory dump). Use REISUB sequence for safe emergency reboot.

Q260 How do you check kernel module dependencies?

Answer 'modinfo modulename' shows module info including dependencies (depends field).
:
'modprobe --show-depends modulename' shows all modules that would be loaded.
'depmod -a' rebuilds module dependency database in /lib/modules/\$(uname -r)/modules.dep. 'lsmod | grep modulename' checks if loaded.
'/sys/module/modulename/' contains loaded module parameters. 'cat /proc/modules' lists modules with size and usage count.

Q261 What are kernel namespaces in detail?

Answer Seven namespace types: PID (process IDs — container PID 1), Network (network interfaces, routes, iptables), Mount (filesystem mounts), UTS (hostname and domain name), IPC (message queues, semaphores, shared memory), User (UID/GID mappings — rootless containers), Cgroup (cgroup root — added in kernel 4.6). Create: 'unshare --net --pid --fork bash'. Containers typically use all 7. 'cat /proc/PID/ns/*' shows namespace memberships.

Q262 How do you configure kernel parameters at boot vs runtime?

Answer Boot-time (persistent): edit /etc/default/grub GRUB_CMDLINE_LINUX, then
:
'update-grub'. Runtime (temporary): 'sysctl -w param=value' or 'echo value > /proc/sys/path/param'. Runtime (persistent): add to /etc/sysctl.conf or /etc/sysctl.d/99-custom.conf, apply with 'sysctl -p /etc/sysctl.d/99-custom.conf'. Boot params affect hardware and early kernel init; sysctl changes kernel runtime behavior (networking, memory management, security).

Q263 What is the kernel OOM killer?

Answer OOM (Out Of Memory) killer activates when system has no more memory to allocate. It
:
selects and kills processes based on oom_score (higher = more likely to be killed). 'cat /proc/PID/oom_score' shows a process's score. 'echo -1000 > /proc/PID/oom_score_adj' protects critical process from OOM kill (-1000 = never kill). 'echo 1000 > /proc/PID/oom_score_adj' makes it highest priority for killing. Check: 'dmesg | grep -i "out of memory"'.
:

Q264 How do you use kexec?

Answer kexec loads a new kernel without full hardware reinitialization — very fast kernel switch
:
(seconds vs minutes). 'kexec -l /boot/vmlinuz --initrd=/boot/initrd.img --reuse-cmdline' loads a kernel. 'kexec -e' executes the loaded kernel immediately (bypasses BIOS/UEFI). Used by: kdump (crash kernel), fast kernel updates in VMs and cloud, A/B kernel testing. 'systemctl kexec' triggers kexec reboot through systemd. Not suitable for hardware problems.

Q265 What is the difference between interrupt and exception?

Answer : Interrupts: asynchronous signals from hardware (keyboard input, disk I/O complete, network packet received). CPU pauses current task, runs interrupt handler (ISR), then resumes. Hardware interrupts use IRQ lines. Software interrupts (INT instruction) are synchronous. Exceptions: synchronous events caused by CPU while executing (division by zero, page fault, invalid instruction). Types: faults (recoverable – page fault), traps (after instruction – breakpoints), aborts (fatal – double fault).

Q266 How does CPU scheduling work in Linux?

Answer : Linux uses CFS (Completely Fair Scheduler) as default for normal tasks. CFS uses a red-black tree sorted by virtual runtime – gives each process proportional CPU time based on priority (nice value). Real-time scheduling: SCHED_FIFO (fixed priority, runs until blocks or higher priority), SCHED_RR (round-robin among same priority RT tasks). 'chrt -f 50 command' sets RT FIFO priority 50. 'schedtool -r -p 50 PID' changes scheduling. 'ps -eo pid,cls,pri,ni,cmd' shows scheduling class.

Q267 What is memory overcommit in Linux?

Answer : Memory overcommit allows allocating more virtual memory than physical RAM+swap. Controlled by 'vm.overcommit_memory': 0 (default, heuristic – allows moderate overcommit), 1 (always allow – never fail malloc), 2 (never overcommit – strict, total commit $\leq$ RAM+swap * overcommit_ratio). 'vm.overcommit_ratio=50' (used when mode=2). View committed memory: 'grep Committed /proc/meminfo'. Mode 1 used by some VM managers; mode 2 for safety-critical systems.

Q268 How do you enable and use transparent huge pages?

Answer : Transparent Huge Pages (THP) automatically uses 2MB pages instead of 4KB when beneficial. Check: 'cat /sys/kernel/mm/transparent_hugepage/enabled' ([always] always, [madvise] opt-in, [never] disabled). Set: 'echo madvise > /sys/kernel/mm/transparent_hugepage/enabled'. Persistent: add to /etc/rc.local or systemd unit. Databases (Oracle, MongoDB) recommend disabling THP: 'echo never > ...'. Monitor: 'grep -i huge /proc/meminfo'. Benefits: reduces TLB misses for large allocations.

Q269 What is a CPU affinity and how do you set it?

Answer : CPU affinity pins a process to specific CPU cores. 'taskset -c 0,1 command' starts command bound to CPUs 0 and 1. 'taskset -cp 0,1 PID' sets affinity for running process. 'taskset -p PID' shows current affinity as bitmask. 'numactl --cpunodebind=0 command' binds to NUMA node. In systemd: CPUAffinity=0 1 in [Service] section. 'isolcpus=2,3' kernel parameter isolates CPUs from scheduler (for real-time tasks). Use for latency-sensitive or HPC workloads.

Section 10: Advanced Administration & Exam Prep

Questions 271 – 300

Q271 What is the difference between rsync and scp for file transfers?

Answer : rsync: incremental, only transfers changed blocks within files (delta), preserves permissions/timestamps/symlinks with '-a', supports compression '-z', bandwidth limiting '--bwlimit=1000', delete removed files '--delete', dry run '-n'. Best for synchronization and large/repeated transfers. scp: simple, always transfers full files, no delta, encrypted. Best for one-time copies. For large directories with frequent syncs, rsync is dramatically faster after first transfer.

Q272 How do you configure and use SSH agent forwarding?

Answer : SSH agent forwarding passes your local SSH keys to remote servers so you can hop between servers without copying private keys. Enable: 'ssh -A user@server' or set 'ForwardAgent yes' in ~/.ssh/config for a host. On remote server, \$SSH_AUTH_SOCK is set. Security caution: root on the remote server can use your forwarded agent – only forward to trusted servers. Prefer 'ProxyJump' (ssh -J bastion host) for bastion hosts instead.

Q273 What is the difference between TCP and UDP in practical terms?

Answer : TCP practical characteristics: reliable delivery (retransmits lost packets), ordered (reassembles out-of-order), connection overhead (3-way handshake), built-in flow/congestion control. Good for: file transfer, web, email, SSH. UDP practical: no guarantee, no ordering, no connection, minimal overhead. Good for: real-time (VoIP, gaming, video streaming), DNS (fast query/response), DHCP, NTP, SNMP. TCP can be 10-40% slower than UDP for real-time streams due to retransmit delays.

Q274 How do you configure advanced firewall rules with iptables?

Answer : Rate limiting: 'iptables -A INPUT -p tcp --dport 22 -m limit --limit 5/min -j ACCEPT'. Log and drop: 'iptables -A INPUT -p tcp --dport 23 -j LOG --log-prefix "TELNET:" && iptables -A INPUT -p tcp --dport 23 -j DROP'. Stateful: 'iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT'. Port knocking possible with recent module. Save: 'iptables-save > /etc/iptables/rules.v4'. Load on boot: iptables-persistent package.

Q275 What is a certificate authority and how do you create one?

Answer Create a private CA: 'openssl genrsa -out ca.key 4096' (CA private key). 'openssl req -x509 -new -nodes -key ca.key -sha256 -days 3650 -out ca.crt' (self-signed CA cert). Sign a CSR: 'openssl x509 -req -in server.csr -CA ca.crt -CAkey ca.key -CAcreateserial -out server.crt -days 365 -sha256'. Distribute ca.crt to clients and add to trusted store. 'update-ca-certificates' (Debian) or 'update-ca-trust' (RHEL) installs CA system-wide.

Q276 How do you set up passwordless sudo for automation?

Answer In /etc/sudoers.d/deploy (via visudo -f): 'deploy ALL=(ALL) NOPASSWD: /usr/bin/systemctl restart myapp, /usr/bin/rsync'. Restrict to specific commands only — never NOPASSWD: ALL for automation accounts. Set proper file permissions: 'chmod 440 /etc/sudoers.d/deploy'. Test: 'sudo -l -U deploy' lists allowed commands. For Ansible, typically add deploy user to a group and grant group sudo. Always use service accounts, not personal accounts, for automation.

Q277 What is the difference between NFS v3 and v4?

Answer NFSv3: stateless, UDP/TCP, separate MOUNT protocol, requires portmap/rpcbind, no built-in security (uses AUTH_SYS with UID/GID). NFSv4: stateful, TCP only, single port 2049 (firewall-friendly), integrated MOUNT protocol, compound operations (reduced round-trips), ACL support, Kerberos authentication (krb5, krb5i, krb5p for privacy), file delegation (client-side caching). NFSv4 is preferred for new deployments, especially over firewalls. Mount v4: 'mount -t nfs4 server:/path /mnt'.

Q278 How do you implement log centralization?

Answer Options: rsyslog with @/@@remote-server, systemd-journal-remote (journal native), ELK stack (Elasticsearch+Logstash+Kibana), Grafana Loki, Graylog, Fluentd/Fluentbit. rsyslog forwarding in /etc/rsyslog.conf: '*. * @@logserver:514' (TCP). rsyslog server: '\$ModLoad imtcp', '\$InputTCPServerRun 514'. Secure with TLS: use gtls driver. Rotate logs on clients to prevent local disk fill. Tag logs with hostname for source identification.

Q279 What is AIDE and how do you configure it?

Answer AIDE (Advanced Intrusion Detection Environment) monitors file integrity. Install: 'apt install aide'. Config: /etc/aide/aide.conf — define which files to monitor and what attributes. Initialize database: 'aide --init' creates /var/lib/aide/aide.db.new. Copy: 'cp /var/lib/aide/aide.db.new /var/lib/aide/aide.db'. Check for changes: 'aide --check' compares current state to database. Schedule daily: add to cron or systemd timer. Report changes to administrator.

Q280 How do you perform a system backup with rsync?

Answer `'rsync -avz --delete /source/ /backup/'` synchronizes source to backup. `'-a'`: archive mode (recursive, preserves permissions, timestamps, owner, group, symlinks). `'-v'`: verbose. `'-z'`: compress. `'--delete'`: removes files in destination not in source. `'--exclude='.cache'` excludes patterns. `'--bwlimit=10000'` limits bandwidth (KB/s). `'rsync -avz -e "ssh -p 2222" /source/ user@backup-server:/backup/'` remote backup over SSH.

Q281 What is the difference between SNAT and DNAT in iptables?

Answer SNAT (Source NAT): changes source IP of outgoing packets. Used for internet sharing (masquerade). `'iptables -t nat -A POSTROUTING -o eth0 -j SNAT --to-source 1.2.3.4'` or `'-j MASQUERADE'` for dynamic IP. Happens in POSTROUTING chain. DNAT (Destination NAT): changes destination IP of incoming packets. Used for port forwarding. `'iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to-destination 192.168.1.10:80'`. Happens in PREROUTING chain.

Q282 How do you configure VLANs for network segmentation?

Answer Load module: `'modprobe 8021q'`. Create VLAN interface: `'ip link add link eth0 name eth0.100 type vlan id 100'`, `'ip addr add 192.168.100.1/24 dev eth0.100'`, `'ip link set eth0.100 up'`. Persistent (NetworkManager): `'nmcli connection add type vlan con-name eth0.100 dev eth0 id 100 ip4 192.168.100.1/24'`. Each VLAN gets its own subnet. Switch port must be configured as 802.1Q trunk. Useful for network segmentation without additional hardware.

Q283 What is the purpose of umask and how does it affect default permissions?

Answer umask removes permission bits from new file/directory creation. Files start at 666 (no execute), directories at 777. umask 022 removes write from group and others: files get 644 (666-022), directories get 755 (777-022). umask 027: files get 640, directories get 750 (removes group write, all other perms). `'umask'` shows current. Set in `/etc/profile` (system-wide) or `~/.bashrc` (user). Services often set umask in systemd unit: `UMask=0077`.

Q284 How do you configure automatic security updates?

Answer Debian/Ubuntu: `install 'unattended-upgrades'`. Configure `/etc/apt/apt.conf.d/50unattended-upgrades` to enable security-only updates. `/etc/apt/apt.conf.d/20auto-upgrades` enables and schedules. `'unattended-upgrade --dry-run'` tests. RHEL/CentOS: `install 'dnf-automatic'`. Configure `/etc/dnf/automatic.conf`: `'apply_updates = yes'`. Enable: `'systemctl enable --now dnf-automatic.timer'`. Always configure email notifications and reboot behavior carefully in production.

Q285 How do you debug a slow-starting service?

Answer `'systemctl status service'`: check for errors. `'journalctl -u service -b'`: logs from current boot. `'systemd-analyze blame'`: shows startup time of each unit. `'systemd-analyze critical-chain service.service'`: shows dependency chain with timing. `'systemd-analyze plot > boot.svg'`: visual boot timeline. `'strace -T -p $(systemctl show -p MainPID service | cut -d= -f2)'`: trace system calls. Check if service waits on network: `After=network-online.target needed?`

Q286 What is the difference between RAID 5 and RAID 6?

Answer RAID 5: distributed parity across all disks, can tolerate 1 disk failure, minimum 3 disks, usable capacity (N-1) disks. During rebuild after disk failure, if another disk fails, data is lost — rebuild can take hours on large disks. RAID 6: dual parity (two independent parity blocks), can tolerate 2 simultaneous disk failures, minimum 4 disks, usable capacity (N-2) disks. Slower write performance than RAID 5 (two parity calculations). RAID 6 preferred for large disk arrays where rebuild time is long.

Q287 How do you configure SSH ProxyJump for bastion hosts?

Answer SSH ProxyJump routes SSH through an intermediate host. Command: `'ssh -J bastion-user@bastion.example.com user@internal-server'`. Config file (`~/.ssh/config`): `'Host internal HostName 10.0.1.50 User admin ProxyJump bastion.example.com'`. Multiple hops: `'ProxyJump bastion1,bastion2'`. More secure than agent forwarding — private keys never leave local machine. `'ProxyCommand'` is the older equivalent: `'ProxyCommand ssh -W %h:%p bastion'`.

Q288 What is the nftables syntax and how does it differ from iptables?

Answer nftables is modern, consolidated replacement. `'nft list ruleset'` shows all rules. `'nft add rule ip filter input tcp dport 22 accept'`. vs iptables: single command handles IPv4/IPv6/ARP (no separate ip6tables). Tables and chains defined explicitly. Native set/map support. Cleaner syntax. Example: `'nft add table inet filter; nft add chain inet filter input { type filter hook input priority 0; }; nft add rule inet filter input tcp dport 22 accept'`. firewalld uses nftables backend on RHEL 8+.

Q289 How do you set up log rotation for a custom application?

Answer Create `/etc/logrotate.d/myapp`: `'/var/log/myapp/*.log { daily rotate 30 compress delaycompress missingok notifempty create 0640 myapp myapp copytruncate postrotate systemctl reload myapp >/dev/null 2>&1 || true endsript }'`. Key options: daily/weekly/monthly (frequency), rotate N (keep N old logs), compress (gzip), delaycompress (don't compress most recent rotated), copytruncate (copy then truncate — for apps that can't reopen log files), postrotate/endscript (commands after rotation).

Q290 How do you configure resource control with systemd slices?

Answer : systemd slices organize units into hierarchy for resource control. Default slices: system.slice, user.slice, machine.slice. Create custom: 'myapp.slice' with [Slice] CPUQuota=50% MemoryMax=2G. Assign services: in service unit [Service] Slice=myapp.slice. 'systemctl set-property myapp.service CPUQuota=30%' changes at runtime. 'systemd-cgtop' shows slice resource usage. 'systemctl status myapp.slice' shows hierarchy. Based on cgroups v2.

Q291 What are common LFCS exam topics to focus on?

Answer : Essential Operations (25%): basic commands, man pages, help, file management. User/Group Management (15%): useradd, usermod, sudo, PAM. Networking (20%): ip command, nmcli, firewall, SSH, DNS. Storage (25%): LVM, filesystems, mount, RAID, quotas. Service Management (15%): systemctl, systemd units, cron, logging. Key skills: troubleshooting (not just configuration), understanding WHY things work, reading documentation, combining tools with pipes and scripts.

Q292 How do you create a bootable USB drive in Linux?

Answer : 'dd if=/path/to/iso of=/dev/sdX bs=4M status=progress' writes ISO to USB. Always confirm target device with 'lsblk' before running — dd will destroy data on wrong device. 'sync' after dd to ensure all data is written. Alternative: 'cp ubuntu.iso /dev/sdX' works for hybrid ISOs. 'mkusb' and 'balenaEtcher' provide GUI. For UEFI bootable: use 'isohybrid' or 'Ventoy' (multi-ISO support). Never use the wrong target device.

Q293 How do you interpret /proc/net/tcp?

Answer : /proc/net/tcp shows raw TCP socket info. Columns: sl (slot), local_address (hex IP:port), rem_address (remote hex IP:port), st (state: 01=ESTABLISHED, 0A=LISTEN), tx_queue:rx_queue, timer info, uid, inode. Convert hex: '0100007F:0016' = 127.0.0.1:22. State codes: 01=ESTABLISHED, 02=SYN_SENT, 06=TIME_WAIT, 0A=LISTEN. Use 'ss' or 'netstat' for human-readable output — /proc/net/tcp is raw but useful for scripting.

Q294 What are the steps to recover a forgotten root password?

Answer : 1) Reboot and access GRUB menu (hold Shift). 2) Edit kernel line: press 'e'. 3) Find 'ro quiet splash' and change to 'rw init=/bin/bash' (or add 'rd.break' for RHEL). 4) Press Ctrl+X to boot. 5) If SELinux: 'mount -o remount,rw /sysroot; chroot /sysroot; touch /.autorelabel'. 6) 'passwd root' to change password. 7) 'sync; reboot -f'. Alternative: boot from live media, mount root filesystem, chroot, then passwd.

Q295 How do you use cron for complex scheduling?

Answer Complex cron syntax: '0 9-17 * * 1-5 /cmd' runs hourly 9AM-5PM weekdays. '* /30 * * * * /cmd' every 30 minutes. '0 0 1,15 * * /cmd' on 1st and 15th. '0 0 * * 0 /cmd' every Sunday. Use MAILTO="" to disable emails. Use >> to append logs. Full environment: add PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin at top of crontab. 'run-parts /etc/cron.daily' executes all scripts in a directory – used by system cron.

Q296 What are the differences between soft limits and hard limits for users?

Answer Soft limits: current enforced limit, user can raise up to the hard limit value. Set: 'ulimit -S' or in /etc/security/limits.conf with 'soft'. Hard limits: maximum ceiling, only root can increase. Set: 'ulimit -H' or with 'hard' in limits.conf. Example limits.conf: '@developers soft nproc 5000' and '@developers hard nproc 10000'. 'ulimit -a' shows current soft limits. Users can freely decrease soft limits. Applications that need high limits must set them before dropping privileges.

Q297 How do you configure logwatch for automated log reporting?

Answer logwatch analyzes system logs and sends daily email reports. Install: 'apt install logwatch'. Config: /usr/share/logwatch/default.conf/logwatch.conf and override in /etc/logwatch/conf/. Key settings: Output=mail, MailTo=admin@example.com, Detail=Low/Med/High, Range=yesterday. Run manually: 'logwatch --detail High --mailto admin@example.com --range today'. Schedule via cron (usually pre-configured in /etc/cron.daily/). 'logwatch --service sshd' reports only SSH.

Q298 What are POSIX ACLs and how are they different from standard permissions?

Answer Standard permissions have three classes (owner, group, other) – limited to one user and one group. POSIX ACLs add fine-grained entries: named users, named groups, and a mask. 'setfacl -m u:alice:rw,u:bob:r,g:devs:rw file' grants specific access. 'getfacl file' shows all entries including mask. Mask restricts maximum effective permissions for ACL entries. ACL-enabled files show '+' in 'ls -l' output. Requires filesystem support ('acl' mount option or default on ext4/xfs).

Q299 How do you tune the network stack for high performance?

Answer Key network sysctl tunings: 'net.core.rmem_max=134217728' (max TCP receive buffer). 'net.core.wmem_max=134217728' (max TCP send buffer). 'net.ipv4.tcp_rmem=4096 87380 134217728'. 'net.core.netdev_max_backlog=250000'. 'net.ipv4.tcp_congestion_control=bbr' (use BBR algorithm). 'net.ipv4.tcp_fastopen=3' (reduce handshake latency). 'net.core.somaxconn=65535' (listen backlog). Apply: 'sysctl -p'. Verify: 'ss -s' and performance testing with iperf3.

Q300 What is your approach to systematic Linux troubleshooting?

Answer : Structured approach: 1) Define the problem precisely — what's broken, when started, what changed. 2) Gather data: logs (journalctl, /var/log/), dmesg, system status (top, df, free, ss). 3) Form a hypothesis. 4) Test the hypothesis (check one thing at a time). 5) Fix the root cause, not just symptoms. 6) Verify the fix. 7) Document the solution. Key principle: change one thing at a time. Use version control for config files. Always have a rollback plan. Don't panic — work methodically.